

Alternative test methods, declaration of capacity and test of heat pumps in multiple climates zones

Nordic Council of Ministers



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A Nordsyn study on heat pumps in the context of Ecodesign and Energy labelling

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Contents

Authors 2

Foreword 4

Executive summary 5

Scope 7

Test method: Number of test points and defrost evaluation 8

Background 8

Analysis of overlapping test scheme 9

Summary - Overlapping test scheme 19

Recommendations - Overlapping test scheme 20

Analysis of air to water heat pumps defrosting behavior 21

Summary - Defrosting behavior 26

Recommendations - Defrosting behavior 27

Low capacity declaration 28

Air to air heat pumps 28

Air to water heat pumps 33

Summary - Low capacity and colder climate declaration 36

Recommendations - Low capacity and colder climate declaration 37

Alternative test method (compensation method) 38

Background - RISE Experience 38

Difference between the compensation method and other alternatives/solutions and advantages/disadvantages 39

Summary - Alternative test method 43

Recommendations - Alternative test method 44

Appendix A 45

About this publication 50

Foreword

The study presented in this report has been performed on behalf of the Swedish Energy Agency within the Nordic cooperation Nordsyn, sponsored by the Nordic Council of Ministers. The study was performed by RISE Research Institutes of Sweden AB. Any opinions set out in the study are those of RISE and do not necessarily reflect the opinions of the Nordsyn members.

Emma Olsson, The Swedish Energy Agency

Executive summary

This study consists of three parts covering different topics related to heat pump evaluation and declaration. For each part, results of investigations and analysis are presented and suggestions for improvement regarding Ecodesign and Energy labelling regulations, or other regulations/standards, for different kind of heat pumps are presented.

The three parts are:

1. Test method: Number of test points and defrost evaluation
2. Low capacity declaration
3. Alternative test methods (compensation method)

Test method: Number of test points and defrost evaluation

For air to air heat pumps it would be possible to implement an overlapping test scheme for average and colder climate. This can be performed by a requirement that P_{design} at colder climate is 46% higher than P_{deisgn} at average climate. This proposal limits the costs of testing since one additional test point is required to include colder climate in one test sequence.

Another alternative for air to air heat pumps would be to implement a test method as the SP-method 5323 where performance data is interpolated/extrapolated to perform seasonal performance evaluations on different heat load curves. This method is based on testing one heat pump with a number of limited climates and values for P_{design}. With 7–9 test points the method can calculate the SPF for three model houses in three cities.

For air to water and brine(water) to water heat pumps with a varying water temperature outlet at each test condition it would be more complicated to achieve an overlapping test scheme.

According to market survey, one third of the air to air heat pumps seem to use an overlapping test scheme since their P_{design}-ratio between average and colder climate is close to the right percentage.

The evaluation period is sometime too short to include defrost period. When the test time is extended to include a defrost it is showed that COP is for most cases 6–11% higher for a steady state test compared to if a defrost was included. The difference is larger for an outdoor temperature of 2 or 7°C in comparison to -7 och -12°C. The impact on the SCOP value is highly dependent on the heat pump, if a longer testing time makes the evaluation period include the defrost for 2 of the test points, the impact on SCOP is estimated to up to 5%.

Low capacity declaration

Both air-to-air heat pumps and air to water heat pumps are commonly declared with low P_{design} for the average climate, meaning they will likely have high SCOP.

Based on a market investigation on 96 air to air heat pumps, from 13 different

manufacturers, with 36 being declared for both the average and the colder climate, we see that all the latter are declared at lower P_{design} for the average climate than for the colder climate. This means that P_{design} for the average climate is low, likely the reason is to achieve better Energy labeling values. To our knowledge this is common for air to water heat pumps as well.

RISE proposal is to restrict the manufacturers to market heating capacities that are higher than P_{design}. That would effectively reduce the low capacity declaration as the heating capacities used in marketing are more important than P_{design} on the Energy labeling.

Alternative test methods (compensation method)

The compensation method enables a test that includes the control system of the heat pump. In comparison, the current method is based on locked frequencies where the control used in real life environment is overruled. Hence, the test with the method with locked frequencies may provide performance results that differ from the heat pump performance under real life conditions. There is also a risk that the test method described in EN 14511:2018 and EN 14825:2018 with locked frequencies gives small or no incentive to the manufacturers to develop efficient and stable control functions (with small temperature fluctuations) of the heat pumps.

In a comparison between the methods it was noted that:

- 3 of 4 heat pumps showed up to 10% lower SCOP values with the compensation method
- The largest difference in COP between the methods was for the lowest part load conditions

With the compensation method there is a risk that the design of the testing rig has a non-negligible influence on the results. However, according to our experience, this influence is very small.

Using the compensation method may take longer time and be more costly to perform in comparison with a locked frequency test, especially for heat pumps with poor control. But at the same time, the compensation method allows for a more automated test that could shorten the total test time and could also decrease the necessary man hours. It is also important to add requirement on stable operation of the heat pump in the regulation to reduce the risk of long an expensive testing times, but more important to ensure high thermal comfort in real-life operation.

Before applying the compensation method, the first step should be to establish the method fully. Our suggestion is a Round Robin test, deep analysis of the results and revision of the method. In our opinion, the eco-design/energy class limits should not be changed when introducing the compensation method. The reason is that the current levels are mainly based on RISE early measurements using the compensation method

Scope

This study consists of three parts. For each part, results of investigations and analysis are presented and suggestions for improvement regarding Ecodesign and Energy labelling regulations, or other regulations/standards, for different kind of heat pumps are presented.

The analysis is done for the relevant types of products: i) air conditioners/air-air heat pumps (lot 10) and ii) heat pump space heaters (lot 1).

1. Test method: Number of test points and defrost evaluation

Investigation of possible synergies of testing heat pumps for more than one climate zone at the same time. The study has investigated simplified methods for test/ calculation of efficiency in multiple climates. This encompass, analysis of only adding test points for the colder/warmer climate zone, without redoing the overlapping test points (medium climate) or by calculations and/or extrapolations. Other barriers for an increased use of the voluntary labelling information in cold climate are also presented. The study also evaluates if the heat pumps defrosting behavior, and the energy use associated with it, in real use is reflected in standardized laboratory tests.

2. Low capacity declaration

The study assesses to what extent suppliers declare a lower capacity in order to improve the declared seasonal performance, for one or more climate zones, and what the implications are. Investigation and suggestions on how the incentives for suppliers to declare their products for non-representative low capacities could be decreased.

3. Alternative test methods (compensation method)

> Investigation and analysis of the appropriateness of alternative test methods for test of heat pumps (compensation test method). The differences between the compensation method and the method described in EN 14511:2018 and EN 14825:2018 are analysed including evaluation of advantages and disadvantages for the methods.

The study includes assessment of the methods with regards to:

- accuracy and reliability
- comparability between products
- reflection of the heat pumps performance in real life use, including noise
- enabling of independent tests
- costs of testing for market surveillance authorities and for manufacturers

Test method: Number of test points and defrost evaluation

Background

In EN14825:2018 there are three reference design conditions: average (A), warmer (W) and colder (C). For each climate there are defined a reference design temperature (T_{design}) corresponding to a design load (P_{design}) also referred to as Prated in regulation 813/2013 and 811/2013, which is declared by the manufacturer or in some cases determined during third party testing.

Operation limit temperature (TOL) and bivalent temperature (T_{biv}) are declared by the manufacturer, in some cases these temperatures can also be determined during third party testing. At the bivalent temperature the unit is declared to have a capacity able to meet 100 % of the heating load without supplementary heater.

Figure 1, Figure 2, Figure 4 and Figure 5 show part load condition A-E, with the assumption that TOL equals T_{design} for each climate. According to EN14825:2018 TOL are defined:

- for the average heating season, the dry bulb temperature is –7 °C or lower
- for the colder heating season, the dry bulb temperature is –15 °C or lower

The part load of the house has been calculated with the part load ratio given in EN14825:2018, see Table 1. In tables below the part load ratio of the house for the average climate and the colder climate are referred to as Part load ratio AC and Part load ratio CC.

Test condition	Outdoor temperature (°C)	Part load ratio Average Climate	Part load ratio Colder Climate
D	12	0.15	0.11
C	7	0.35	0.24
B	2	0.54	0.37
A	-7	0.88	0.61
E	-10/-22	1	1

Table 1 *Part load ratio at average and colder climate according to EN14825:2018.*

Analysis of overlapping test scheme

In order to achieve equally heat load at test points for average and colder climate, P_{design} at colder climate must be 46% greater than P_{design} at average climate. The 46% difference is derived from the linearization shown in Figure 1 below. For the Figures 1-4 P_{design} at colder climate equals 10 kW and P_{design} at average climate equals 6.8 kW.

Today, there is no requirement that P_{design} for colder climate needs to be larger than P_{design} at average climate. For some manufacturers P_{design} has been declared as below, to show there is no overall principle how to declare P_{design}.

- P_{design} at colder climate is higher than P_{design} at average climate
- P_{design} at colder climate is lower than P_{design} at average climate
- P_{design} at colder climate is equal to P_{design} at average climate

There are different options to determine P_{design} and T_{biv} along with the requirements in EN14825:2018 clause 6, for example:

1. The manufacturer chooses to first define P_{design} and calculate the required heating load at part load conditions. The bivalent temperature can be determined as the cross section of the heat load curve and the heat pump curve.
2. The manufacturer chooses to first define T_{biv} and determine the heating capacity, then calculates P_{design}.

Air to air heat pumps

For an air to air heat pump the indoor temperature is 20°C at each test point. The variables are the heating demand of a fictious house at different outdoor temperatures. This makes it possible to interpolate test points when the test points coincide into a straight line, shown in Figure 1.

The dotted line in Figure 1 show the operation of a fictive air to air heat pump. When testing an air to air heat pump it has been noticed that some heat pumps cannot operate at the demanded part load of the house at condition D (12°C). In real installation this means that the heat pump would operate on/off. The heat pump is assumed not be monovalent.

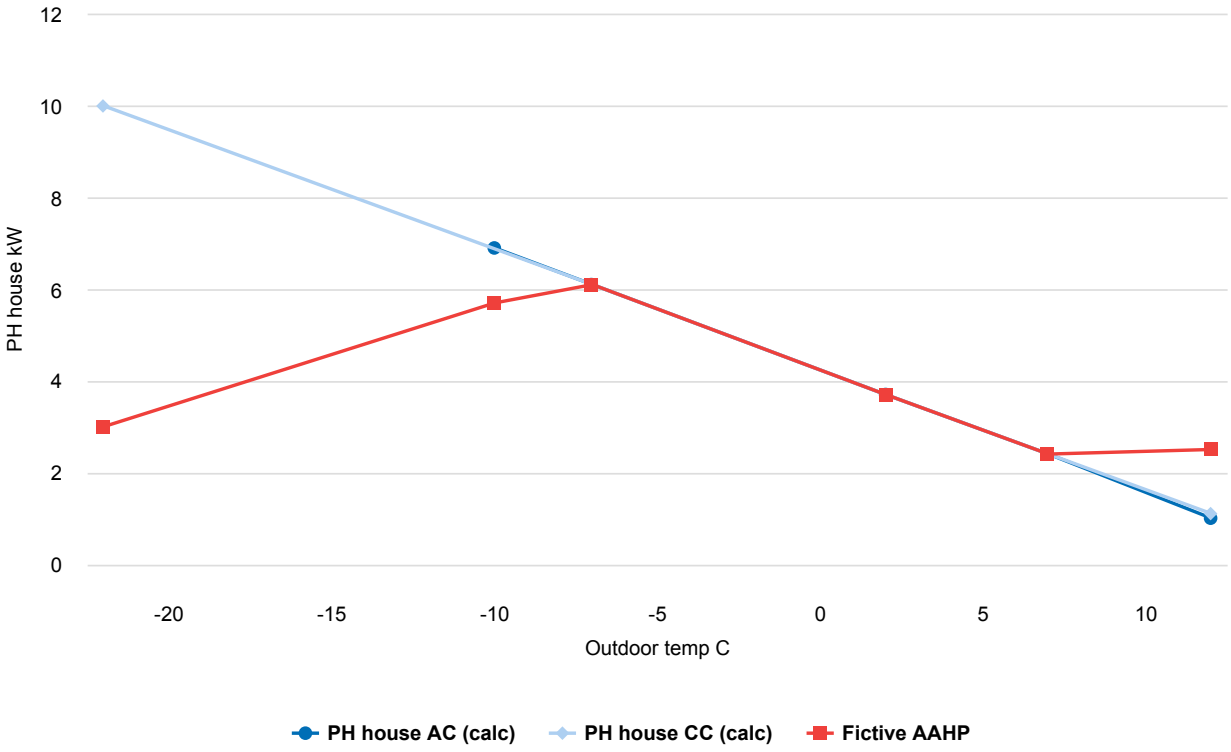


Figure 1 Air to air heat pump *– Heating demand of the house for average and colder climate.*

Note: *Test points A-E acc. EN14825*
P_{design} *AC = 6.9 kW*
P_{design} *CC = 10 kW*

In Figure 2 the bin hours for the average and colder climates are added. For outdoor temperature between -30°C and -23°C the bin hours are 0. For colder climate the bin hours start at -22°C and for average climate at -10°C. The outdoor temperature with the highest number of bin hours are for colder climate at 1°C and for average climate at 3°C.

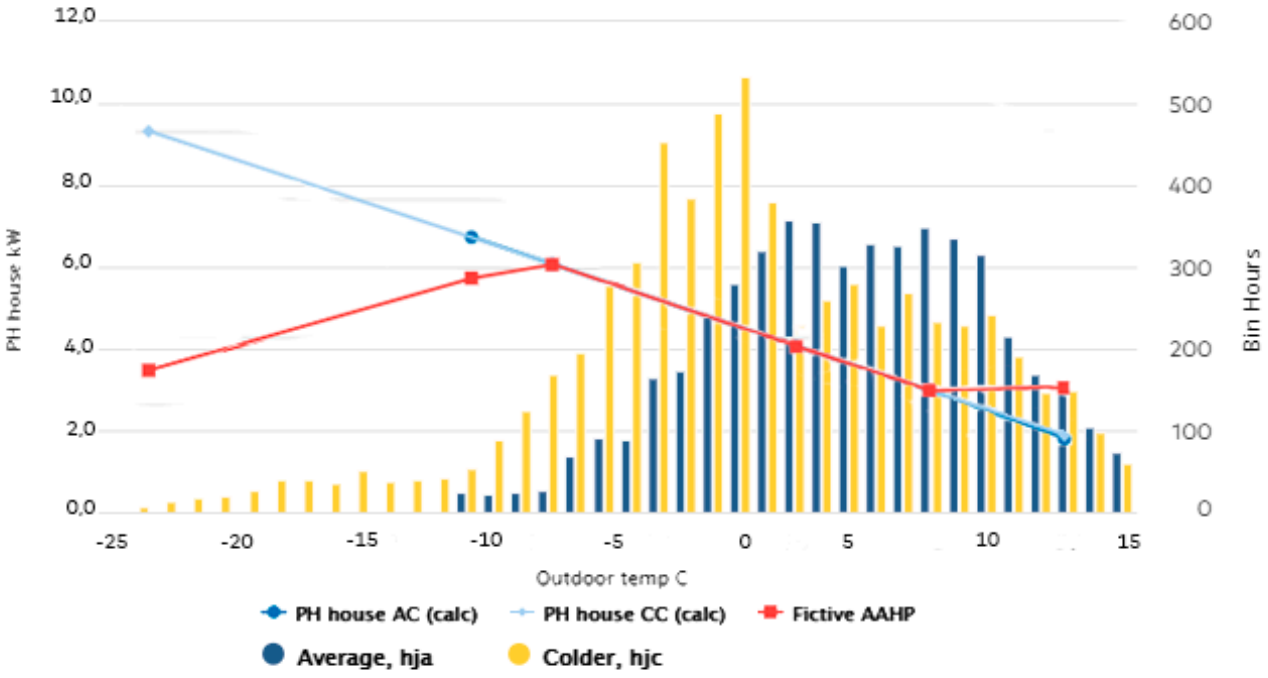


Figure 2 Air to air heat pump *– Heating demand of the house and applicable climate bins for average/colder climate.*

Note: *Test points A-E acc. EN14825*
P_{design} *AC = 6.9 kW*
P_{design} *CC = 10 kW*

Another option to test air to air heat pump is to use the calculation principle in SP-method 5323, shown in Figure 3. In this method 7 test conditions are required to be tested and 2 additional conditions if needed. With 7-9 test points the method can calculate the SPF for three model houses in three cities. Nine (9) cases in total. The model houses have been defined by their heating demand of 4, 6 and 8 kW at -15°C. Calculations according to SP-method 5323 uses the climate for three Swedish cities, Luleå, Stockholm and Malmö. SP-method 5323 is used for tests in Sweden with Kyl och Värmepump Importörerna, referred to as KVI.

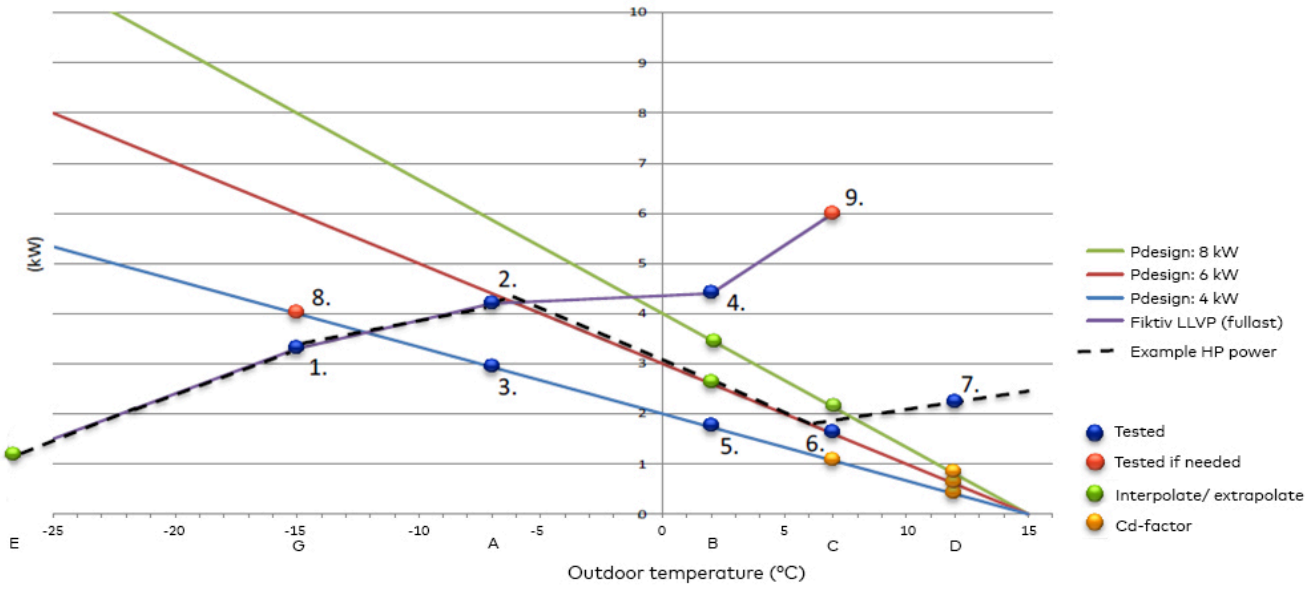


Figure 3 *Calculation of SPF according to SP-method 5323.*

Since the SP-method 5323 Include 3 houses (heat loads) and 3 climates the customer get more and sometimes more relevant information that can be used when choosing a heat pump. Today manufacturers define the Pdesign themselves resulting in a risk that there are few heat pumps that fit for a specific heat load. Hence, the information given on a label might not be relevant for a customer with another heat load than defined by the manufacturer. With three different predefined heat loads the customer can focus on the performance values that matches the customers heat load best.

With EN 14825:2018 up to 18 different test points (normally 12-15 test points) must be tested to calculate a SCOP value for the three different climates. With SP-method 5323, SCOP/SPF values are achieved for three climates and three heat loads with 7-9 test points.

SP-method 5323 can be used both with the compensation method and a method with locked frequencies. When using the compensation method better representation of real-life use compared to a method with locked frequencies is achieved, since the control of the heat pump is included in the test.

Since there are fewer test points on one heat load curve, and a higher degree of interpolation is used, the uncertainty on SCOP/SPF is somewhat higher compared to EN 14825:2018. Our estimation of the uncertainty of SPF is 8% with SP-5223 while the estimated measurement uncertainty for SCOP according to EN14825:2018 is 5%.

Air to water and brine(water) to water heat pumps

Determining the performance of an air to water heat pump or brine(water) to water heat pump more parameters varies during testing,

- Temperature applications at low, intermediate, medium and high with test points at different water outlet temperatures. Most common to test are low and medium temperature application.
- Variable/fixed flow rate and/or temperature difference. For brine to water heat pumps the flow/temperature difference can either be variable or fixed, both on the source and sink side.

With the same requirement as for an air to air heat pump, P_{design} at colder climate (at -22°C) must be 46% greater than P_{design} at average climate (at -10°C) to achieve the same heat load at the test points, but still the temperatures will vary. Figure 4 shows the heat curve of the house, at the defined test point in EN14825:2018 and the water outlet temperature for both average and colder climate and at low and medium temperature application. The air to water heat pump is assumed not to be monovalent and the brine(water) to water heat pump is assumed to be monovalent.

Even though the heat curve for average and colder climate coincides with the outlet temperature on the sink side would vary as shown in Figure 5. This makes it more complicated to interpolate between test points since it is in fact 3 dimensional. The water outlet temperatures are dependent on the declared bivalent temperature and TOL, which are different from heat pump to heat pump.

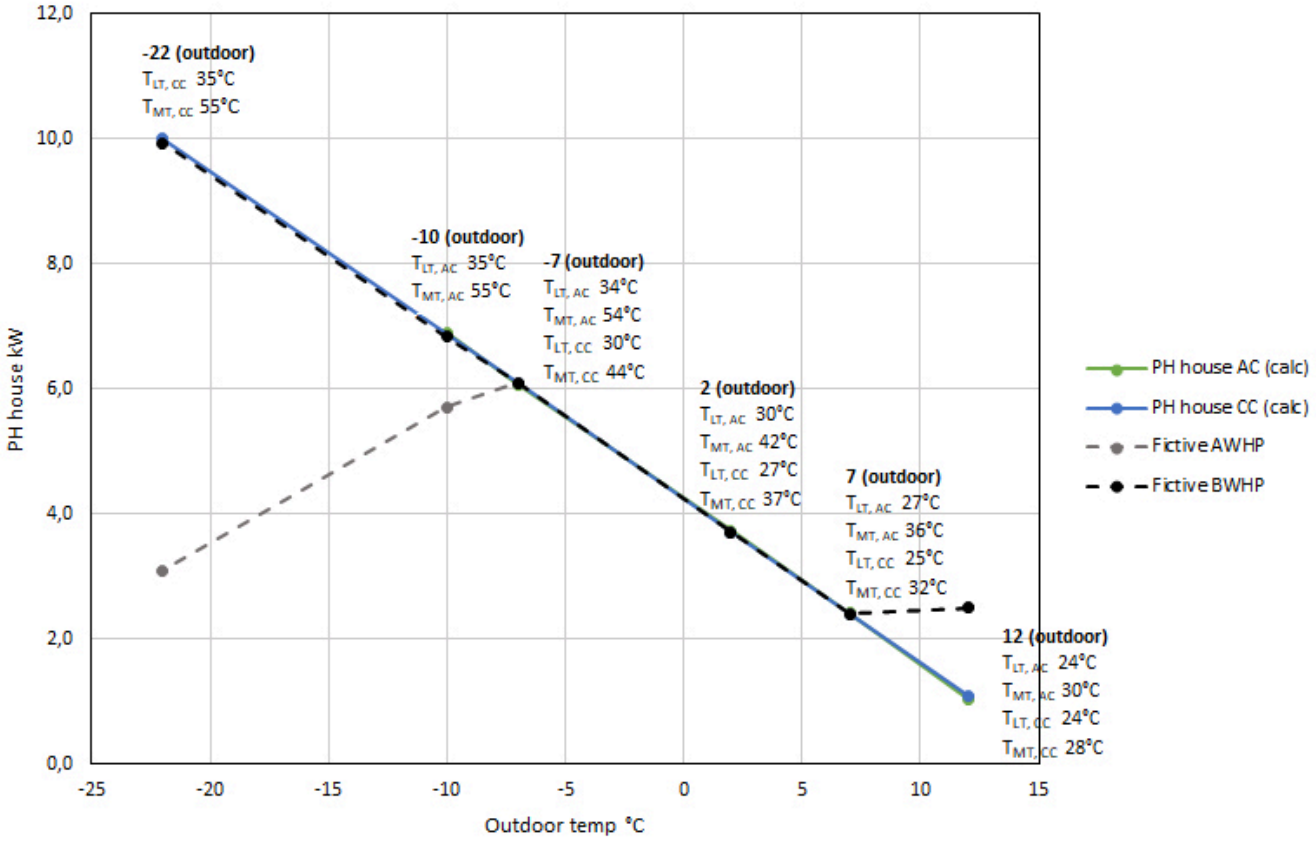


Figure 4 Air to water and brine(water) to water heat pump *– Heating demand of the house for average and colder climate. Figure includes outlet temperature on the sink side at low and medium temperature applications.*

Note: *Test points A-E acc. EN14825*
P_{design} *AC = 6.9 kW*
P_{design} *CC = 10 kW*

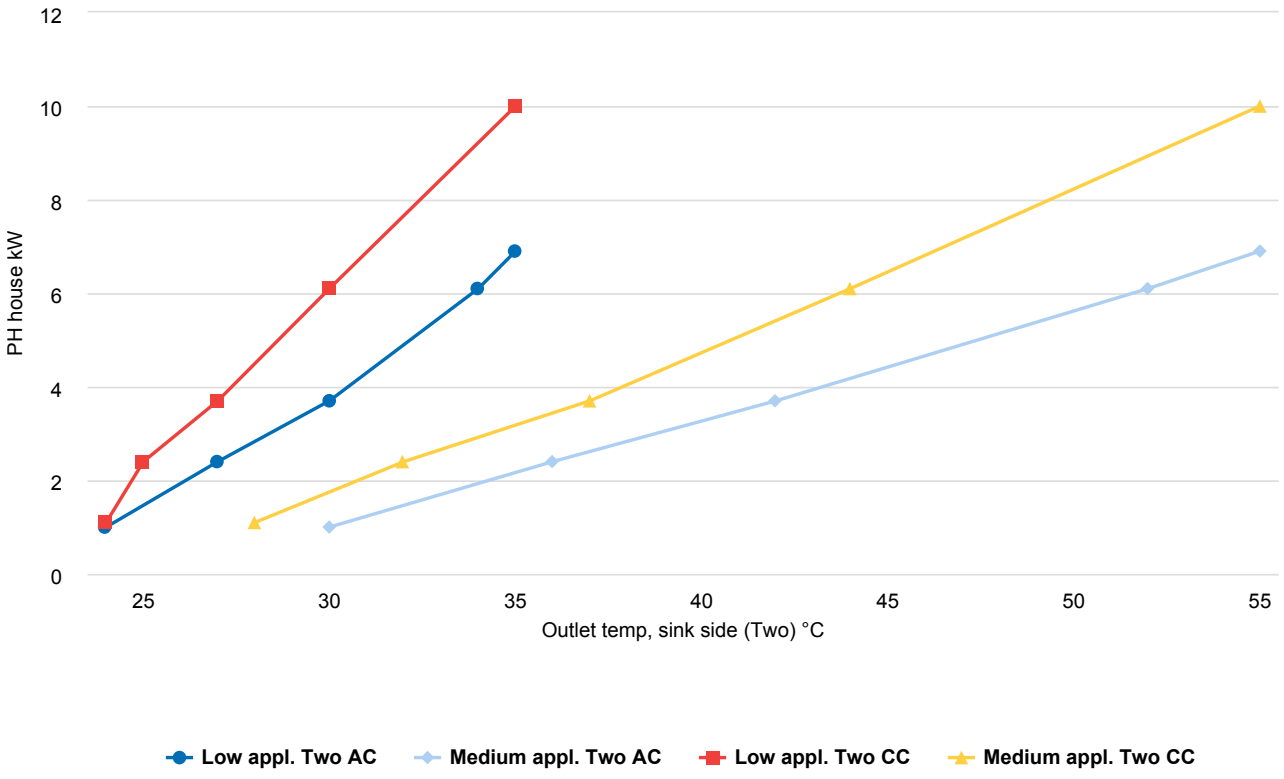


Figure 5 Air to water and brine(water) to water heat pump *– Heating demand of the house for average and colder climate. Figure includes outlet temperature on the sink side at low and medium temperature applications.*

Note: *Test points A-E acc. EN14825*
P_{design} *AC = 6.9 kW*
P_{design} *CC = 10 kW*

According to EN14825:2018 the part load of the house for each test condition is calculated from P_{design}. The calculated part load value is used during performance test to check that the heat pump operates within 10% of the measured heating capacity. If the heat pump cannot operate at lower frequencies during a higher outdoor temperature when the heat pump will operate on/off in a real installation, the test point is compensated with a higher water temperature outlet (according to eq. 35 in EN14825:2018). This will lead to a different supply temperature and a different heating capacity during measurements compared to the standard test points (calculated part load based on P_{design}).

An example of how the temperature and heating capacity can differ from measurements compared to what is calculated according to the standard values is shown in Figure 6 for a specific heat pump. The behavior is not general, it can differ from heat pump to heat pump. The figure shows average and colder climate for medium temperature application for an air to water heat pump.

At test point B, C and D for colder climate the water temperature outlet was compensated and for average climate at test point C and D.

For this test the P_{design} at average climate was defined by the manufacturer to 10

kW and for colder climate to 9.4 kW. The bivalent temperature and TOL for average climate were -10°C and for colder climate -17°C. See Figure 7.

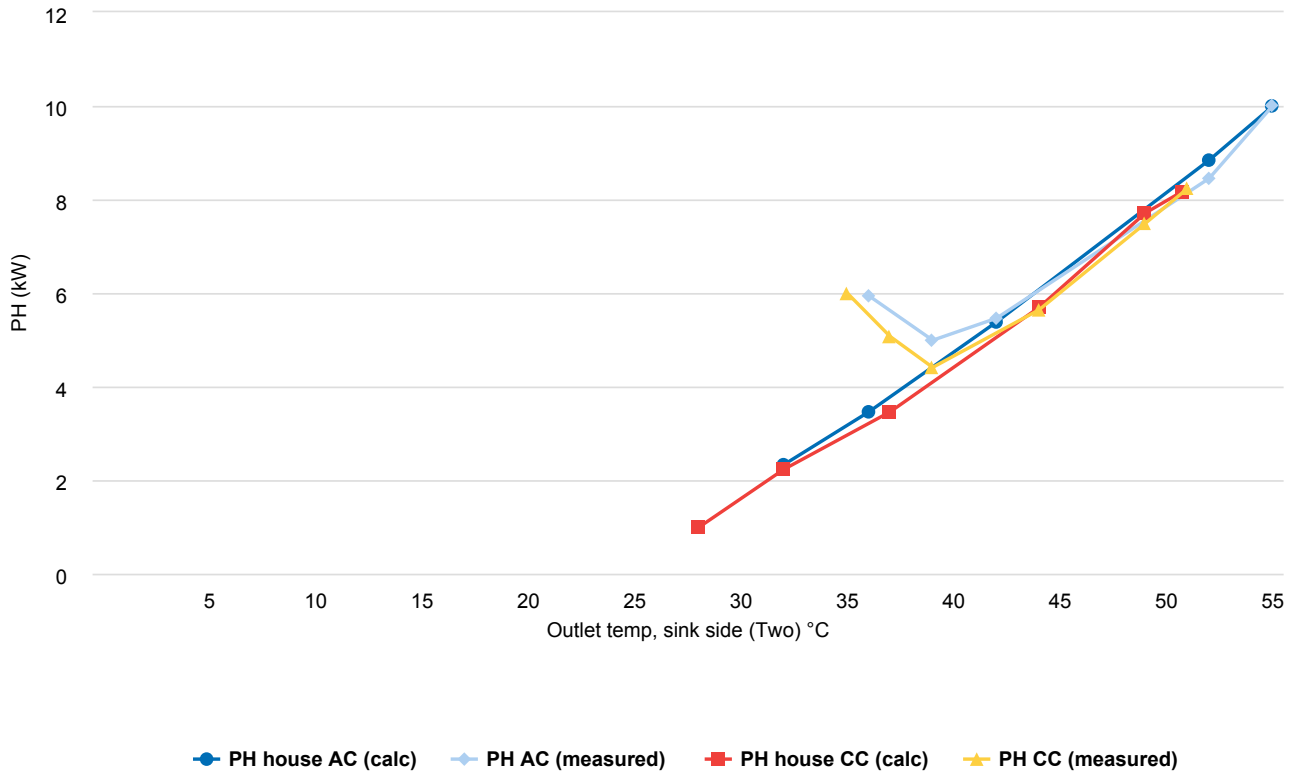


Figure 6 Anonymized heat pump A *– Part load of the house and outlet temperature for average and colder climate at test points according to EN14825:2018, medium temperature application, compared to measured data.*

Note: *Test points A-E acc. EN14825 and measured values*
P_{design} *AC = 10 kW*
P_{design} *CC = 9.4 kW*

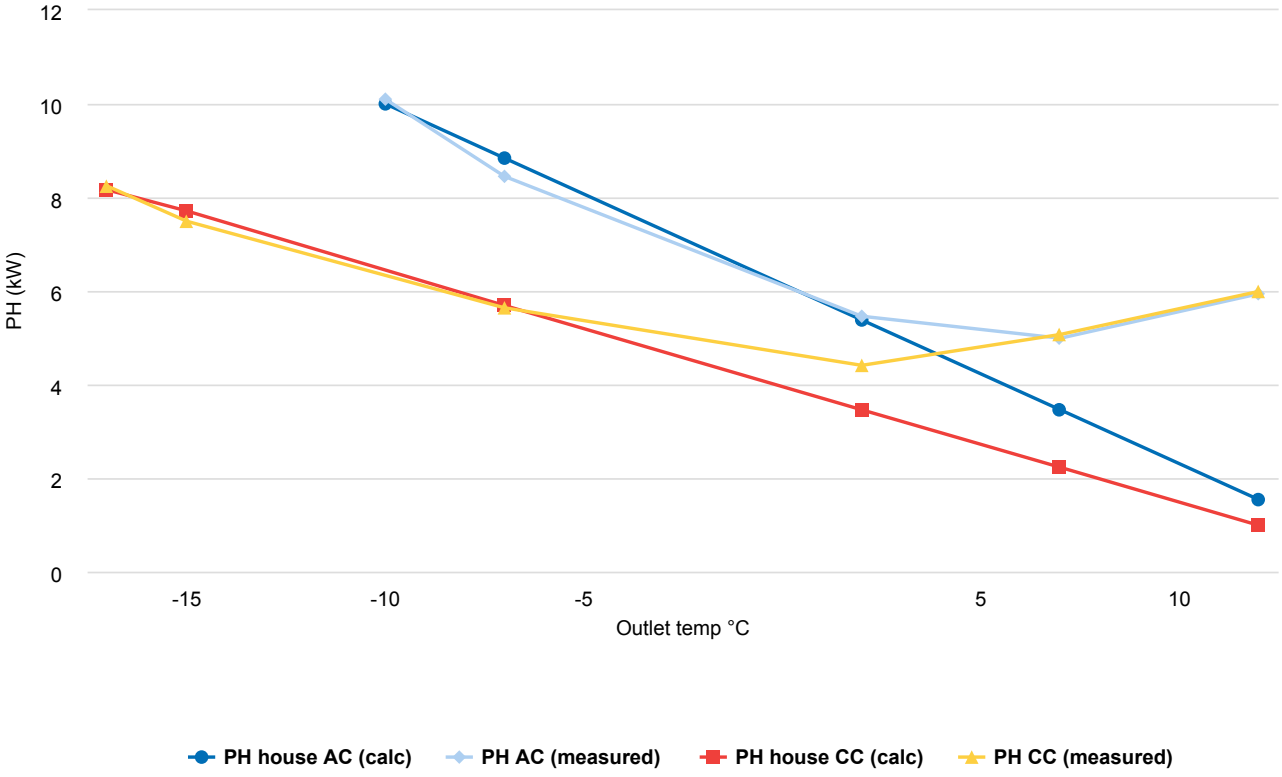


Figure 7 Anonymized heat pump A *– Part load of the house and outdoor temperature for average colder and warmer climate at test points according to EN14825:2018, medium temperature, compared to measured data.*
Note: *Test points A-E acc. EN14825 and measured values*
P_{design} *AC = 10 kW*
P_{design} *CC = 9.4 kW*

With a smaller difference between P_{design} for the climates the outlet temperatures are closer to each other. In this case with a P_{design} for average climate 6% greater than P_{design} at colder climate. See Figure 8. The figure shows only the calculated part load of the house based on P_{design}.

Instead of focusing on achieving an equal heat load for the test points the outlet temperature on the sink side were more linear if P_{design} had a smaller difference between average and colder climate. Thus, the test point shown in Figure 8 will not be linear at a higher outdoor temperature since the heat pump may not operate at these low frequencies and the test point is performed with a higher water temperature outlet (also shown in Figure 6). This will make it more complicated to interpolate when it is not sure before testing how the heat pump will operate during these test points. But if the heat pump operates at the lowest frequency it is showed in Figure 6 and Figure 7 that the measured heating capacity is equal at test point C (7°C) and D (12°C).

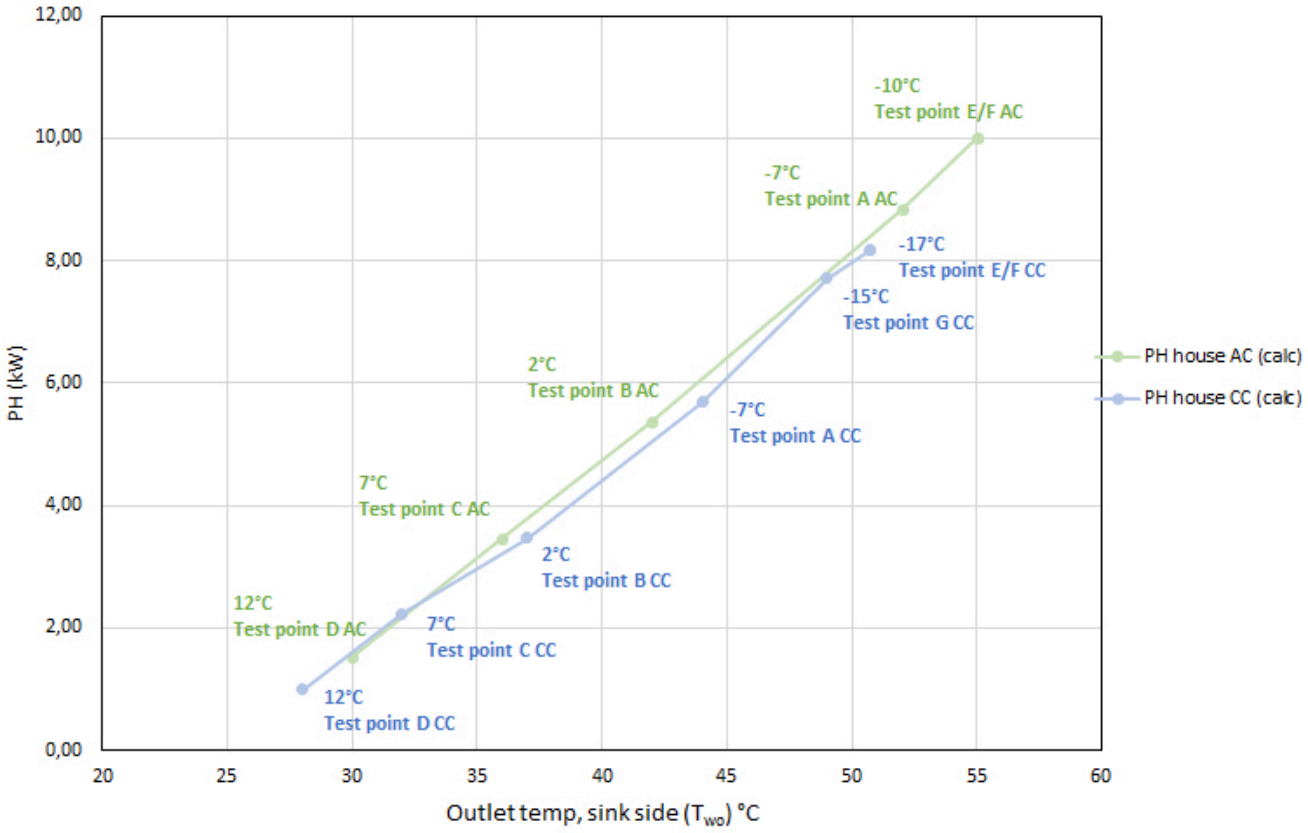


Figure 8 Air to water heat pump *– Part load of the house and outlet temperature for average and colder climate at test points according to EN14825:2018, medium temperature application.*

Note: *Test points A-E acc. EN14825*
P_{design} *AC = 10 kW*
P_{design} *CC = 9.4 kW*

Most of the tests for air to water and brine(water) to water heat pumps performed at RISE in recent years are tested for the Heat Pump Keymark (HP KEYMARK). However, in the Nordic countries a certification for the HP Keymark is not required. In Denmark, HP KEYMARK certified heat pumps can be used to get access to public subsidies based on their full compliance with the ErP regulations. But HP KEYMARK products are currently not recognized by the Danish Energy Agency's list of heat pumps (Varmepumpelisten), which provides homeowners with an overview of energy efficient heat pumps whose performance is documented by third party full SCOP testing; however, the eligibility for subsidies in Denmark is not linked to the Varmepumpelisten in any way.

The HP KEYMARK test scheme includes only a few test points to be tested according to EN14825:2018. For average climate, test points at bivalent temperature and one other part load condition per temperature application is required. For any other climate the bivalent temperature condition is tested. Only testing a few test points, HP KEYMARK have included surveillance tests to control the validity of performance for other models declared in the same subtype.

Summary - Overlapping test scheme

For air to air heat pumps it would be possible to implement an overlapping test scheme for average and colder climate. This can be performed by a requirement that P_{design} at colder climate is 46% higher than P_{deisgn} at average climate. Today, there is no requirement on how P_{design} should be declared. This proposal limits the costs since one additional test point is required to include colder climate in one test sequence.

Another alternative for air to air heat pumps would be to implement a test method as the SP-method 5323 where performance data is interpolated/extrapolated to perform seasonal performance evaluations on different heat load curves. This method is based on testing one heat pump with a number of limited climates and values for P_{design}. With 7-9 test points the method can calculate the SPF for three model houses in three cities. Nine cases in total.

For air to water and brine(water) to water heat pumps with a varying water temperature outlet at each test condition it would be more complicated to achieve an overlapping test scheme. To achieve an equal heat load for average and colder climate at each test condition, P_{design} at colder climate must be 46% higher than P_{deisgn} at average climate. With this requirement the water outlet temperature will be distributed with a large spread as shown in Figure 4. If the difference is decreased between P_{design} at average and colder climate the outlet temperature would be more linear, but then the heat load would not be equal for the testing condition.

Making it further complicated is the operation of the heat pump at part load conditions with higher outdoor temperatures. For these conditions when the heat pump would operate on/off in a real installation, the test is performed with a higher water temperature outlet.

Even though it would be possible to achieve an overlapping test scheme it would require a more advanced calculation tool and it would be more complicated to define in the standard. There is also a risk that more test points are required to include colder climate in one test sequence, which could lead to less profitability.

According to the market survey, see chapter Low capacity declaration, P_{design} for the colder climate declared on air-to-air heat pumps sold in Sweden is on average 35% higher than for the average climate. One third of those heat pumps had near 46% (42-52%) higher Pdesign for the colder climate, meaning the manufacturers are already using the overlapping test scheme.

Recommendations - Overlapping test scheme

For air to air heat pumps it would be possible to implement an overlapping test scheme for average and colder climate. This can be performed by a requirement that P_{design} at colder climate is 46% higher than P_{deisgn} at average climate.

Advantages

- This proposal limits the costs since one additional test point is required to include colder climate in one test sequence.
- One third (30 out of 90) of the heat pumps that are declared for the colder climate on sold by one retailer on the Swedish market are already using the overlapping test scheme.

Disadvantages

- A predefined ratio of the P_{design} for average and colder climate limit the freedom a heat pump manufacturer must define the P_{design} for the colder climate. Hence, there is risk that a heat pump designed for a colder climate and with a high performance in the low temperature range, might not be declared with the high-performance figures as it "deserves".

An alternative method for air to air heat pumps would be to implement a test/calculation method as the SP-method 5323 where performance data is interpolated/extrapolated to perform seasonal performance evaluations on different heat load curves.

Advantages

- With 7-9 test points the method can calculate the SPF for three model houses in three cities and three different climates.
- With three different predefined heat loads and three different climates the customer can focus on the performance values that matches the customers heat load/climate best.

Disadvantages

- The estimated measurement uncertainty of the seasonal performance factor SPF is 8% with SP-method 5323 in comparison of an uncertainty of SCOP of 5% in EN 14825:2018.

For air to water and brine(water) to water heat pumps with a varying water temperature outlet at each test condition it would be very complicated to achieve an overlapping test scheme. Hence, our recommendation is to use the method described in EN 14825:2018.

Analysis of air to water heat pumps defrosting behavior

For air to water heat pumps tested in a laboratory environment according to EN14511:2018 and EN14825:2018 the heat pumps are controlled with an additional computer or some kind of control device, in some cases there is a specific menu in the display panel of the heat pump to set the required frequencies.

The defrost pattern are different from heat pump to heat pump. In some cases, the heat pump defrosts on a timewise pattern. Usually for this kind of defrosting it is not noticed any deviation on the sink side with a dT exceeding 2.5% i.e. decreasing water temperature outlet (T_{wo}) shown in Figure 9. For some heat pumps it is noted that frosting occurs on the outdoor heat exchanger and with a decreased water temperature outlet shown in Figure 10. Some heat pumps with variable flow rate will operate with a higher flow during the defrost period than the heating period.

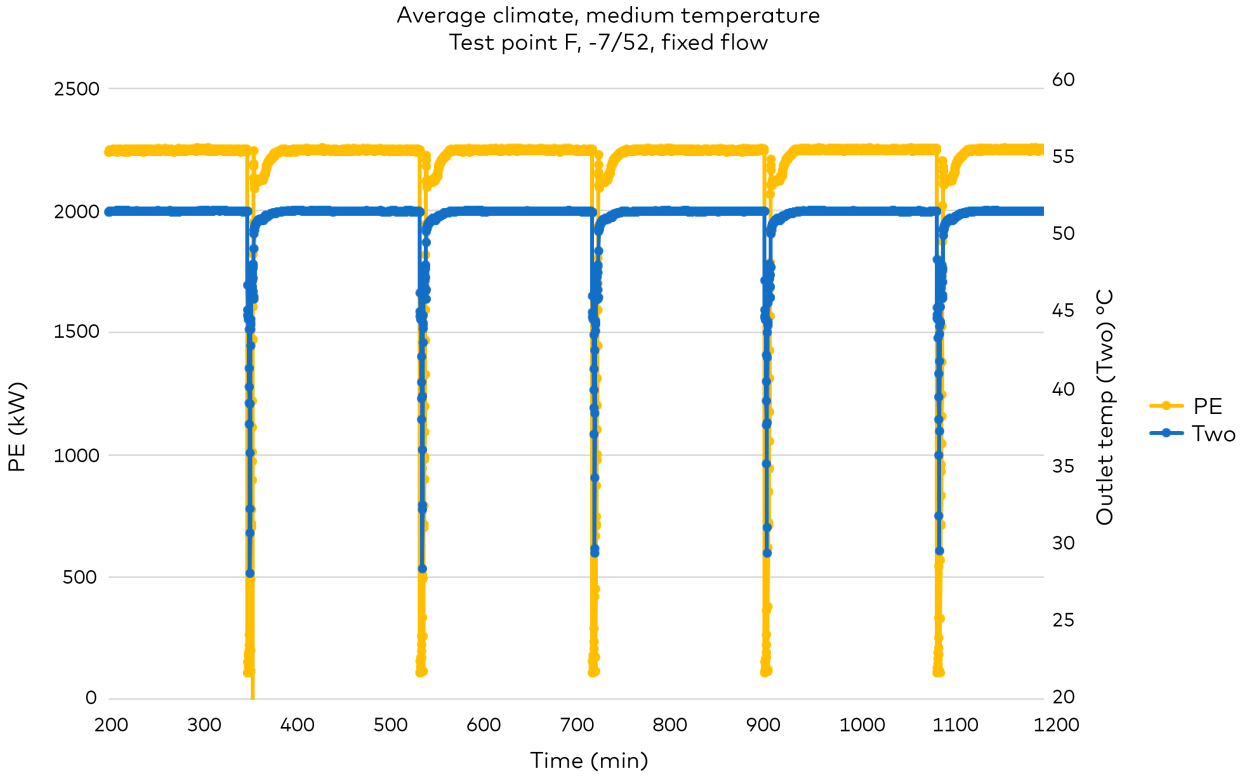


Figure 9 *Measurements of PE (electric power) and temperature out from the heat pump of anonymized heat pump B.*

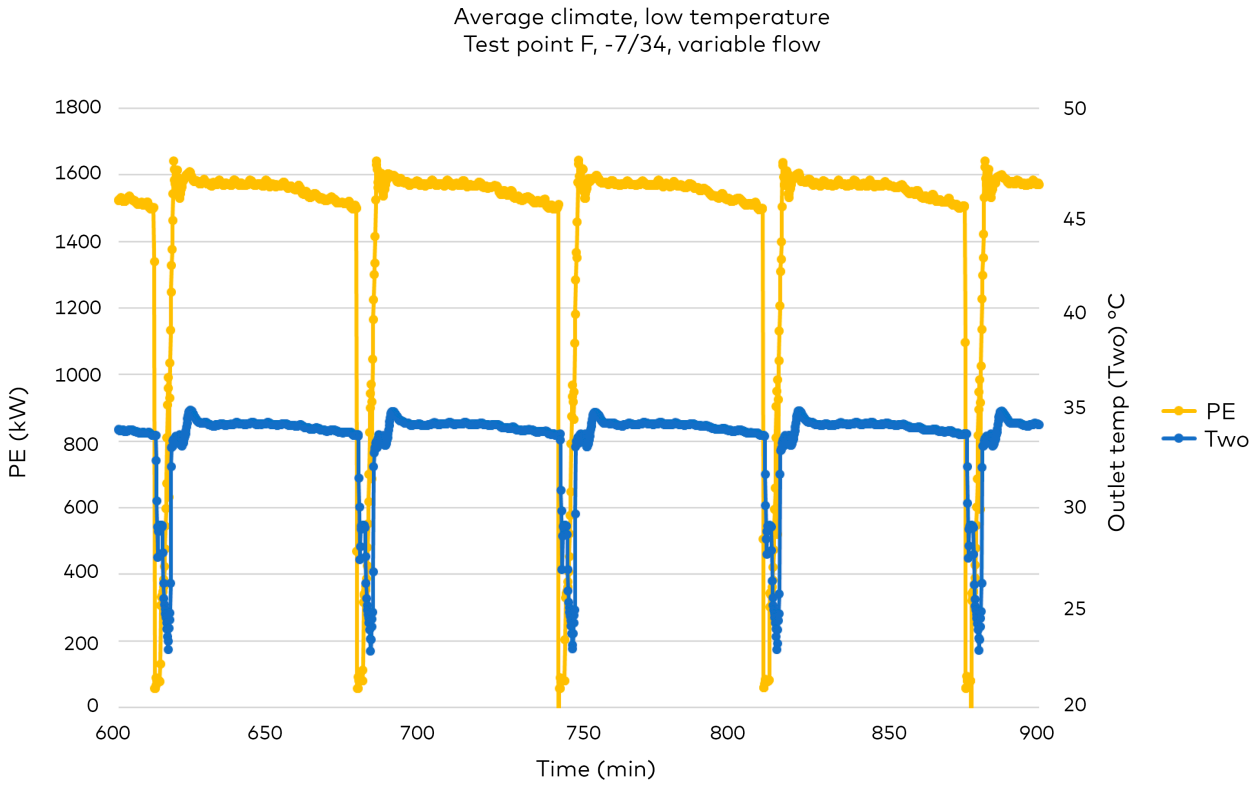


Figure 10 *Measurements of PE (electric power) and temperature out from the heat pump of anonymized heat pump C.*

An example of a clockwise defrost is shown in Figure 9 with defrost every 180 min (Test condition F at average climate and medium temperature application). The test was designated a steady state test since the heat pump did not defrost within the equilibrium period (60 min) or data collection period (70 min) or with a dT exceeding 2.5%. As for Figure 10 the test was designated a transient test since the heat pump defrost during the data collection period. This test would not be in the category of a steady state test since the defrost occurs every 50 min. During the heating period of the test, when the tolerances for steady state condition is fulfilled, the heating capacity is 14% higher than during the transient test and COP is 8% higher.

EN14511:2018 recommend the test to start with a defrost. If the test for heat pump B (Figure 9) would start with a undefined start, after the heat pump operated about 1 hour after a defrost for example, the test would be designated a transient test. Or if hypothetical the dT during the data collection period of 70 min would exceed 2.5% the test would be designated a transient test. Evaluating the test condition in Figure 9 as a transient test, with a data collection period of 180 min, COP is 2% for a steady state test.

For another manufacturer at test condition F, average climate and medium temperature application the duration of the heating period was also 180 min and with no decreased electrical power consumption or temperature outlet like Figure 9.

For the same manufacturer in another test with a different heat pump model the duration of the heating period was 150 min at test condition F, average climate and low temperature. These both exceeds the total of 130 minutes for the equilibrium and the data collection period with no deviation on the water temperature outlet.

Evaluating a heat pump performance is currently discussed within CEN/TC 113/WG 8, the flow chart describing the procedure for determining if a test is transient or steady state can be interpreted differently between laboratories. In working document N 326, N 335 and N 340 some calculation examples of evaluating if the test is transient or steady state have been made for the same heat pump. For the heat pump in document N 335, the heating capacity can differ 18%, from 55.3 kW to 45.1 kW, depending on if the defrost is included in the data collection period or not. The difference is quite large because of the size of the heat pump. For this example, COP was not presented. For a smaller heat pump (N 340) the heating capacity was between 6.9 kW and 7.86 kW, the steady state test was 12% higher than a transient test and COP was 10% higher for a steady state test.

The number of bin hours for test condition F are lower than for test condition B (2°C) that will have more of an impact on SCOP. The bivalent temperature can be declared in a range of -10°C to 2°C for average climate, the range is normally between -10°C and -7°C, in some cases -5° and -6°C. At an outdoor temperature of 2°C the heat pumps behavior is different from heat pump to heat pump. In some case the heat pump can operate more than 23 hours without defrosting. In some case the defrost pattern are as in Figure 9 and Figure 10.

As presented in Nordsyn study on air to water heat pumps in humid Nordic climate[1] the difference in COP is for most cases 6–11% higher for a steady state test compared to if a defrost was included.

1. C. Haglund Stignor, T. Walfridson 2019, *Nordsyn study on air to water heat pumps in humid Nordic climate,* TemaNord 2019:502, ISBN 978-92-893-5972-6

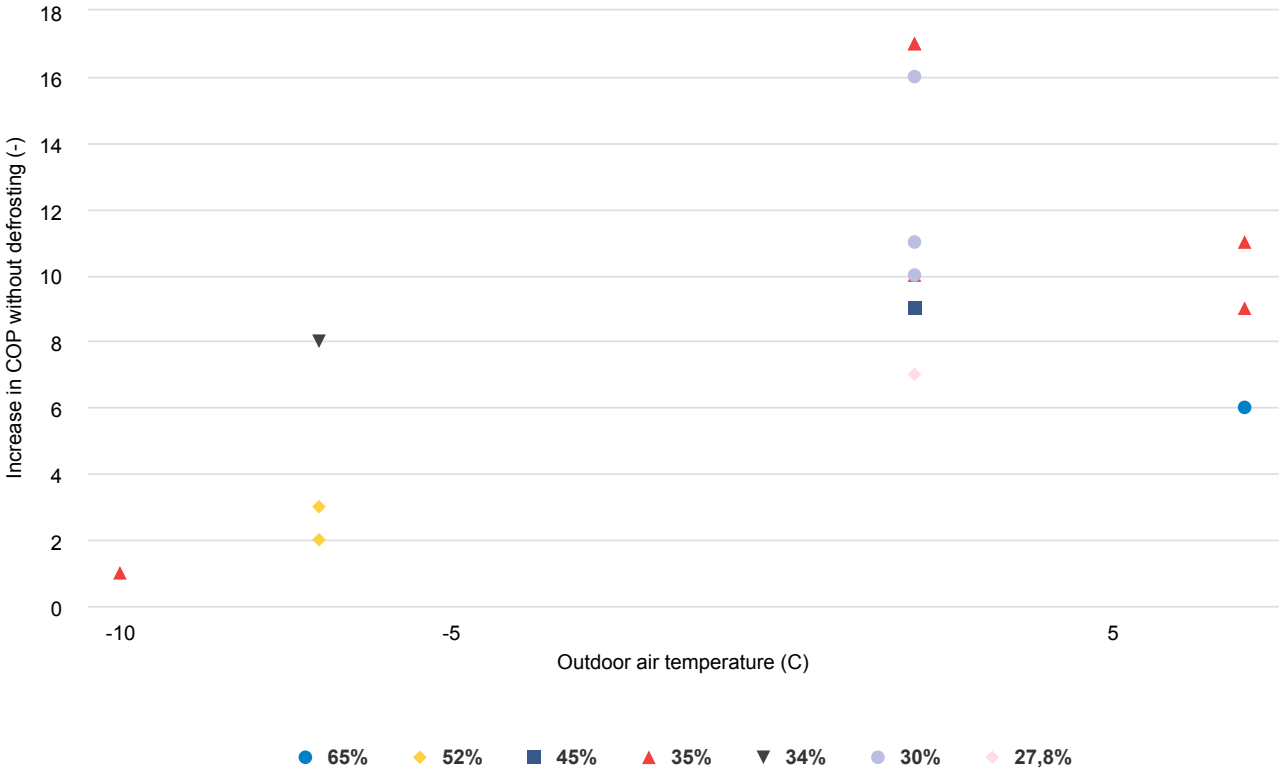


Figure 11 *Increase in COP in laboratory tests of 15 different air to water heat pumps when the heat pump does not defrost within the limited standard test period compared to when the heat pump is let run long enough for defrosting to occur.*

In order to include a defrost in the data collection period it would require longer evaluation periods. But to keep the cost down the evaluation periods are limited to a certain length. In many cases the defrost occur a short time after the equilibrium period (60 min) and data collection period (70 min) ends, if the test starts with a defrost as EN14511:2018 recommend. For some heat pumps the COP difference is rather small if the defrost is included or not, and for some the difference is larger.

As also mentioned in Nordsyn study on air to water heat pumps in humid Nordic climate, there is no proof of a different operation of the heat pump during laboratory tests according to the standard EN14511:2018 and EN14825:2018 compared to real life operation. This is because of the test method allowing the compressor frequency to be fixed.

The compensation method might bring another perspective to how the heat pump operates. In the latest edition of the BAM test guidelines for load-based performance testing[2] of air to water and brine (water) to water heat pumps (dated 2020-06-05), the data collection period for air to water heat pumps have been decreased to 35 min instead of 70 min as stated in 14511:2018. In BAM test method dated (2020-04-09), the data collection period was 180 min.

It is likely that an over-sized (much lower P_{design} than actual capacity) heat pump

2. BAM, S.4 Ecodesign and energy labelling 2020-06-05, *Test guideline for a load-based performance testing, Heat pumps with electrically driven compressors for space heating*

with over-sized heat exchangers can operate for a longer period without having to defrost. The reason is that the frost growth is dependent on temperature, humidity and cooling load of a heat exchanger. And since an oversized heat exchanger can operate with a higher temperature and with a lower specific cooling load, the frost growth is slower in comparison to a normally sized heat exchanger. Hence, a heat pump with better resemblance between the P_{design} and the actual capacity will in the lab test have higher frost growth and will have to defrost more resulting in lower evaluated performance numbers. In the case where this heat pump is installed in a house with higher heat load than the P_{design} (a situation that is likely when there is no regulation controlling this) the SCOP from the lab test will be closer to the real-life performance. For a heat pump that defrost more frequently it is more likely that the defrost is included in the test time that is prescribed in EN 14511:2018 (60+70 min). But in cases where the defrost is occurring after the test time the influence on the result will be higher.

For air to water and air to air heat pumps three steady-state test points normally takes two days to perform. In comparison, transient test points take somewhat longer time. Three transient test points take two or more often three days to perform.

Summary - Defrosting behavior

During laboratory tests according to EN14511:2018 and EN14825:2018 the compressor frequency is set according to the manufacturer's instructions. With these settings it is no proof that the operation of the heat pump is similar to that in a real-life environment. By setting the frequency it is possible that the heat pumps normal control system is overruled (or partially bypassed e.g. in case of the need to defrost). This could be a reason that defrost occurs on a clockwise pattern for some heat pumps with no deviation in measured values before a defrost.

If frosting and defrosting take place for a heat pump at a certain condition in a real installation it will also do so in the standard tests. In many cases the defrost occur a short time after the equilibrium period (60 min) and data collection period (70 min) ends. In order to include a defrost in the data collection period it would require longer evaluation periods. But to keep the cost down the evaluation periods are limited to a certain length. When the test time is extended to include a defrost it is showed that COP is for most cases 6–11% higher for a steady state test compared to if a defrost was included.

Recommendations - Defrosting behavior

RISE recommends usage of an evaluation period of 180 minutes in all cases where:

- a performance degradation is noted over the measurement period,
- or it is noted from ocular inspection, via a camera in the climate chamber, that frost is building up on the outdoor unit heat exchanger.

Advantages

- More heat pumps will be declared with more realistic performance figures since the defrost is included in the lab test.

Disadvantages

- The testing will in some cases take longer time (approximately 10-15% longer) and will therefore be more costly.

Low capacity declaration

Air to air heat pumps

RISE has done an investigation on all 96 heat pumps, from 13 manufacturers, sold by a major Swedish retailor. 47 of these heat pumps had declarations for only the average climate, 36 heat pumps had declarations also for the colder climate, while 13 of the heat pumps had no Energy labeling on the retailors web page. 66 of all heat pumps had a Pdesign of 2,5-2.7 kW or 3.4-3.6 kW, of these 32 heat pumps had declarations for the colder climate. An analysis of the heat pumps show that air to air heat pumps commonly are declared with low P_{design} for the average climate, meaning they will likely have high SCOP, see Figure 12. All of these heat pumps are, for each diagram, declared at the same cooling capacity ($P_{designc}$) meaning they likely, but not necessarily, will have similar heating performance. Despite that they have a wide span in P_{design} and at both climates. On average the P_{design} for the colder climate is 35% higher than Pdesign for the average climate. This indicates that the Pdesign for the colder climate is less misleading compared to the Pdesign at the average climate, but this needs further investigations. On a third of the heat pumps declared for the colder climate the P_{design} was near or very near the 46% needed for the overlapping test scheme described in the chapter *Analysis of overlapping test scheme* above. Note that overlapping test schemes will counteract any try to increase the low capacity declaration.

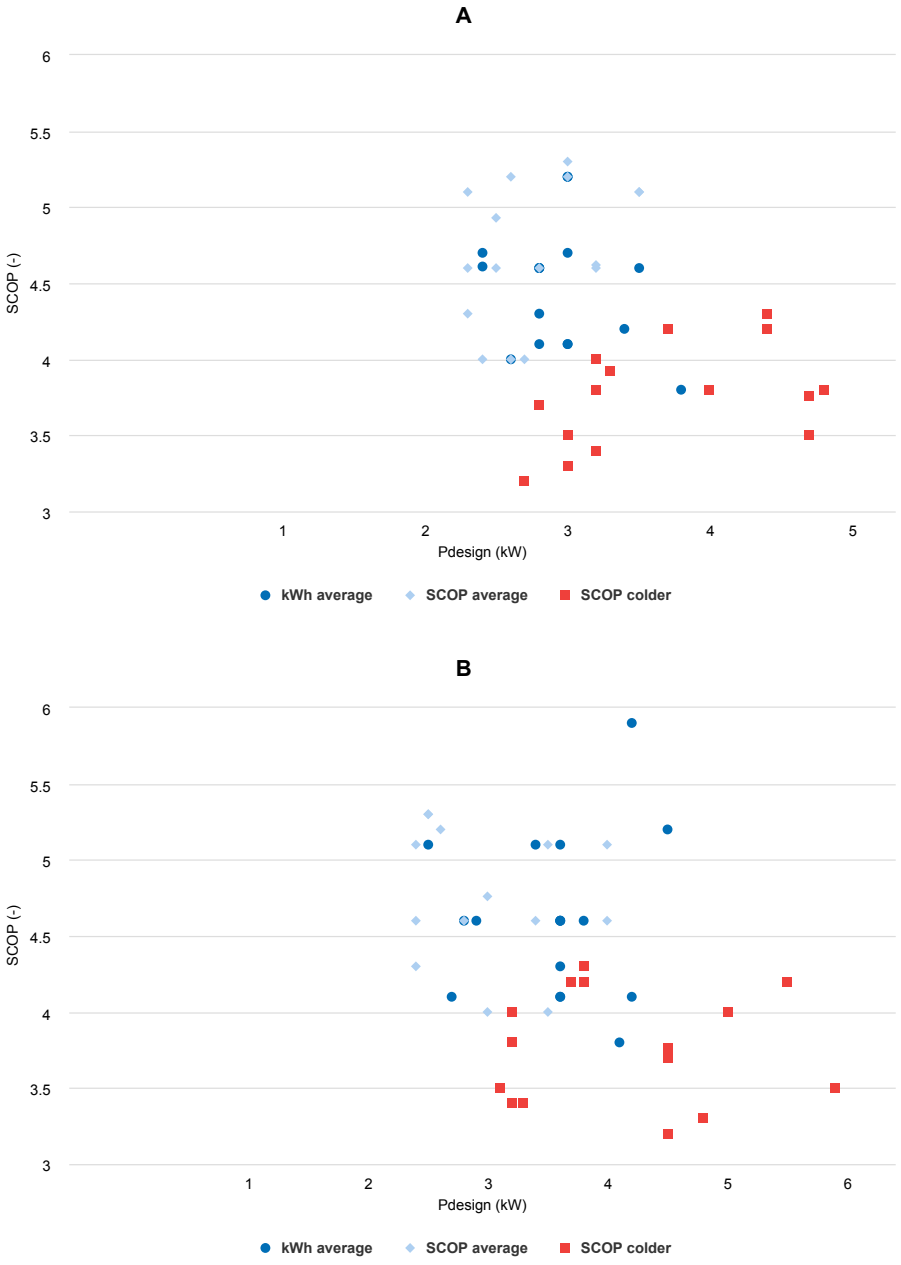


Figure 12 *SCOP as a function of P_{design} for 66 heat pumps of 13 different brands found on a major Swedish retailors web page. All heat pumps that are declared for the colder climate, 32 heat pumps, have higher P_{design} at colder climate than at average climate.* ***A****: Air-to-air heat pumps with a declared cooling capacity of 2.5-2.7 kW.* ***B****: Air-to-air heat pumps with a declared cooling capacity of 3.4-3.6 kW. The "+"-markers shows heat pumps only declared for the average climate.*

As real heating capacity is function of ambient temperature all heat pumps will have heating capacities that in reality are lower at lower ambient temperature. This is due to higher compression ratio (physics) and can not be changed, see Figure 13 below for an anonymized example.

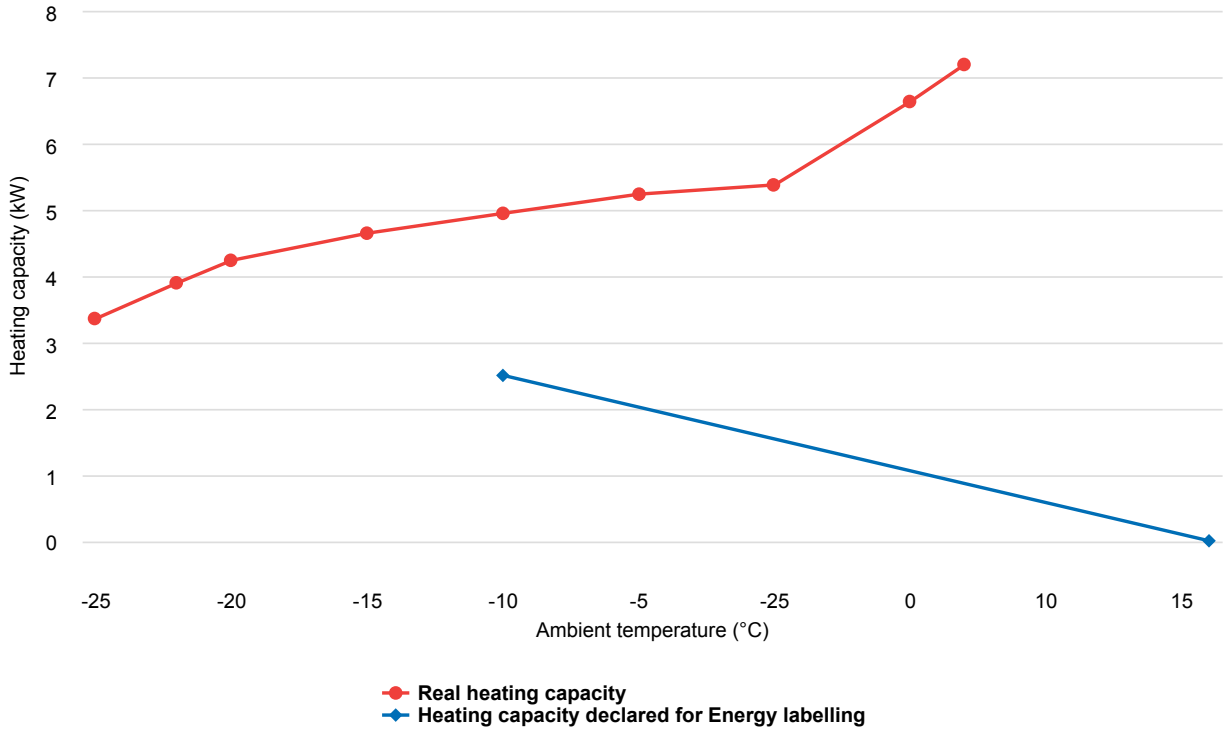
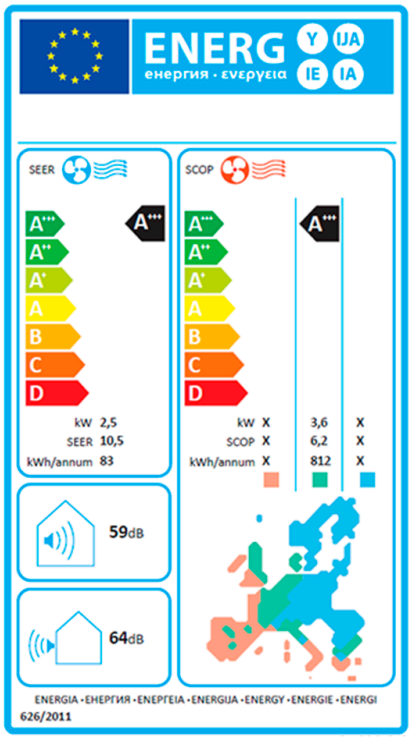


Figure 13 *Real heating capacity (red curve) according to an anonymized heat pump manufacturers openly available data. The real heating capacity at T_{design} for the colder climate (-22°C) is 20% lower than at T_{design} for the average climate (-10°C)*

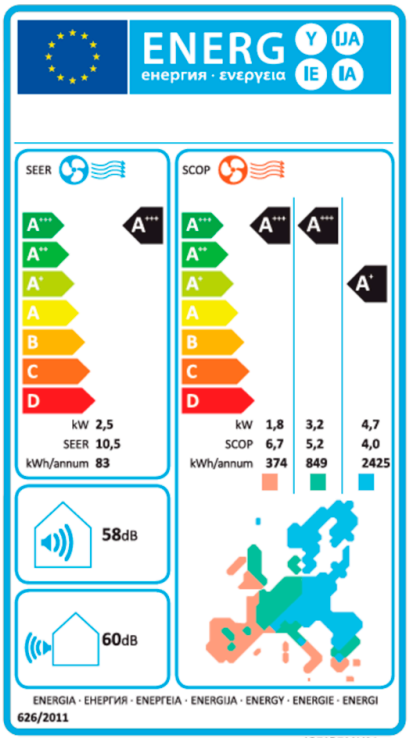
As all heat pumps with declaration for the colder climate shows the opposite, with P_{design} at colder climate being higher than P_{design} at average climate, see Figure 12 above, it can be concluded that all of these are declared at low capacity. In the example in Figure 13 above the P_{design} at average climate (2.5 kW) is half of the real capacity at -10°C (5.0 kW).

Part load COP normally is higher than COP at full load, thus it is beneficial for SCOP to declare the heat pump at low P_{design}. The heat pumps investigated are all declared for the average climate, as it is mandatory, but less than half are also declared for the colder climate, see Figure 12 above. The average climate, especially with low P_{design} will show the customer too high SCOP for the Nordic market (Denmark excluded) and too low power consumption, see three examples in Figure 12.

A



B



C

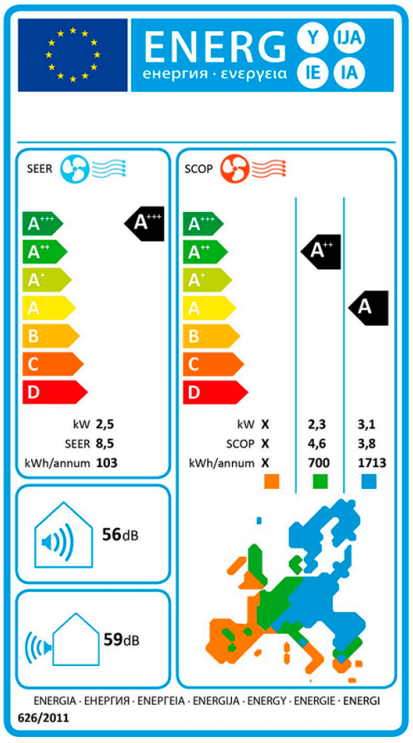


Figure 14 A: *Example of Energy label for an Air to air heat pump declared only for the average climate and at high SCOP, indicating that the* P_{design} *is set low.* ***B and C:*** *Example of Energy label for heat pumps declared for both the average and the colder climates. The SCOP is significantly lower and power consumption is more than twice as high at the colder climate.*

Analysis of the Daikin proposal

Daikin[3] has three proposals to "*to ensure that declared values better reflect real use*":

1. To fix the airflow rates during testing
2. To fix the bivalent point in the calculations of SCOP in average and warm climates
3. To verify that the test settings are within the normal operating range of the unit.

Daikin argues in proposal 1 that the blowing temperature out of the indoor unit (condenser) needs to be high (above 37°C) in order to provide a comfortable climate. This has some bearing for a person being cold, but for a person being warm this is not true. At any given room temperature some people are cold, and some are warm. According to at least Swedish climate and comfort science it is widely know that the main component to comfort is draft, not air temperature. All air movements above 0.15m/s are considered outside the comfort zone (in the heating season). With a well distributed air flow it is thus possible to have higher total air flow through the indoor unit than for an indoor unit with poor air distribution. The argument "*to limit the air flow rate for air to air conditioners/heat pumps to a specific level during testing in order to… ….avoid cold draft effect in heating mode*" could restrict innovation and causing manufacturers with new and innovative air distribution system to use lower air flow than in normal operation. RISE argue that the compensation method should be the first choice in restricting non-normal operation at test. Note that this proposal has no bearing on the low capacity declaration.

APPLiA and Eurovent[4] has the same basic standpoint but problematizes the comfort criteria's more than Daikin does.

All heat pumps already have a bivalent temperature that in reality is lower than T_{design} for the average climate, as concluded in Figure 12 and Figure 13 above. Thus, the argument in proposal 2 of fixing the bivalent temperature to T_{design} is of no benefit to todays Energy labeled heat pumps. Note that this proposal has no bearing on the low capacity declaration.

APPLiA and Eurovent[5] has the same basic standpoint.

In the proposal 3 "*To verify that the test settings are within the normal operating range of the unit*" Daikin propose to fix a lower limit to allowable compressor speed, in order to secure that the heat pumps are not run outside their normal operation limits in tests. This is noteworthy but could also restrict innovation. New and innovative compressor technology enabling lower compressor speeds will effectively be miscredited with this approach. Note that this proposal has no bearing on the low capacity declaration.

3. Daikin, Preliminary comments on the working documents on air conditioners and comfort fans
4. APPLiA and Eurovent, APPLiA and Eurovent position on working documents for air conditioners and comfort fans in view of the Consultation Forum of the 9 September
5. APPLiA and Eurovent, APPLiA and Eurovent position on working documents for air conditioners and comfort fans in view of the Consultation Forum of the 9 September

Analysis of the Svensk Kyl & Värmepumpförening proposal

Svensk Kyl & Värmepumpförening[6] proposes to fixate the P_{design} for heating in the average climate with a factor multiplied by P_{design} for cooling. As P_{design} for cooling can be declared freely this could cause the same problem with low capacity declaration, if the manufacturers in the future will decrease the P_{design} for cooling instead of using the real or near real cooling capacity as they do today. If this is unlikely to happen the Svensk Kyl & Värmepumpförening proposal could be a solution.

Analysis of the EU drafts

The EU drafts[7, 8] has no bearing in the low capacity declaration problem.

RISE proposal

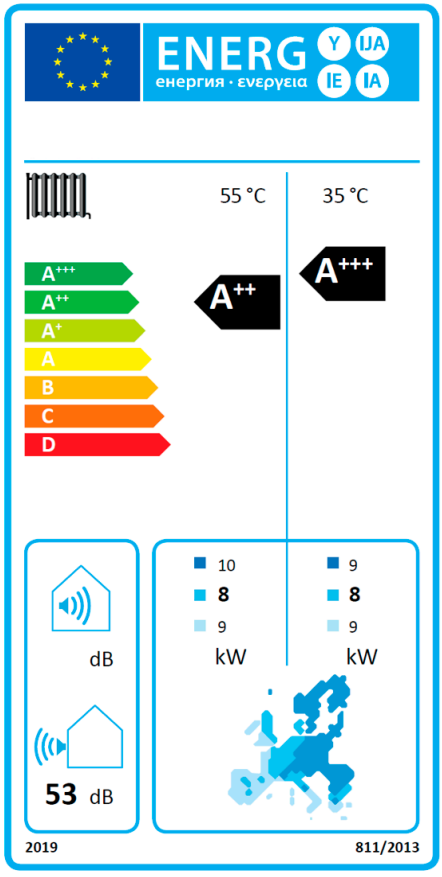
It is a problem that the heat pumps are declared with low capacity as it misleads the consumer by showing SCOP and energy use per annum that seldom will be achieved in a real installation of the heat pump, especially if the heat pump is installed in the colder climate.

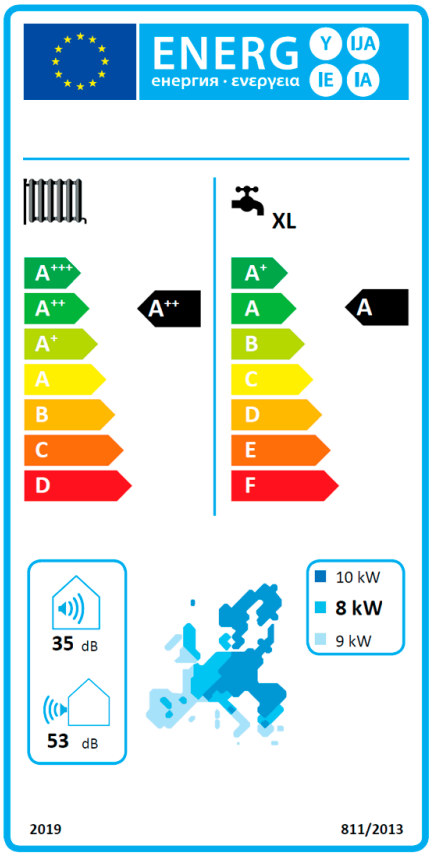
The temperature (T_{design}) at P_{design} is the lowest operating temperature in each climate zone according to the standard. In reality, a very cold winter, that temperature could be surpassed, but likely only a few hours. Using higher heating capacities than P_{design} should be restricted in marketing, as it simply isn't needed according to the declared P_{design}. This approach could effectively reduce the low capacity declarations of air to air heat pumps.

Air to water heat pumps

As with air to air heat pumps the air to water heat pumps are commonly marketed with higher heating capacity than their P_{design}. The example in Figure 14 shows a variable speed heat pump marketed as a 12 kW heat pump, while its P_{design} is in the range of 8-10 kW depending on heating system temperature and climate zone. Note especially that the P_{design} is lower in the average climate than it is in the colder climate despite operating at a higher temperature. It is in reality most likely the opposite, with higher heating capacity at -10°C (average climate P_{design}) than at -22°C (colder climate P_{design}) due to lower refrigerant compression ratio at -10°C

6. Svensk Kyl & Värmepumpförening, Discussion paper regarding Lot 10 products on design heating loads in connection with the energy label
7. COMMISSION DELEGATED REGULATION (EU) .../... of XXX supplementing Regulation (EU) 2017/1369 of the European Parliament and of the Council with regard to energy labelling of air-to-air air conditioners, air-to-air heat pumps and comfort fans repealing Regulation (EU) No 626/2011 with regard to energy labelling of air conditioners
8. ANNEXES to the Commission Delegated Regulation supplementing Regulation (EU) 2017/1369 of the European Parliament and of the Council with regard to energy labelling of air-to-air air conditioners, air-to-air heat pumps and comfort fans repealing Regulation (EU) No 626/2011 with regard to energy labelling of air conditioners





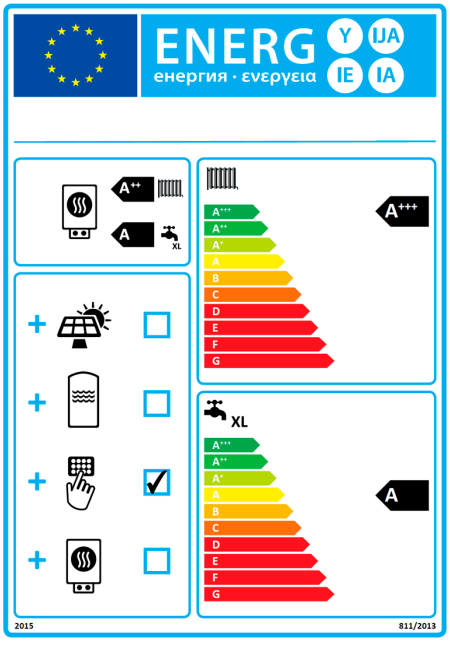


Figure 15 *Example of an anonymous energy labels for a 12kW variable speed air to water heat pump.*

RISE Proposal

Using higher heating capacities than P_{design} should be restricted in marketing, as it simply isn't needed according to the declared P_{design}. Using higher heating capacities in marketing means the heat pump is intended for a different use than the Energy label states. This means that SCOP (shown as energy class on the Energy label) will be lower in reality and that the Energy label is misleading. In the example in Figure 17 above the heat pump should only be marketed as a 10 kW heat pump in the colder climate and 8 kW in the average climate, not as a 12 kW heat pump as it is today.

Summary - Low capacity declaration and colder climate declaration

Both air-to-air heat pumps and air to water heat pumps are commonly declared with low P_{design} for the average climate, meaning they will likely have high SCOP.

Based on a market investigation on 96 air to air heat pumps, with 36 being declared for both the average and the colder climate, we see that all the latter are declared at lower P_{design} for the average climate than for the colder climate. This means that P_{design} for the average climate is low, likely the reason is to achieve better Energy labeling values. To our knowledge this is common for air to water heat pumps as well

The problem of declaring low capacity of heat pumps is not addressed properly in the proposals from Daikin or APPLiA and Eurovent, while Svensk Kyl & Värmepumpförening proposal could be a solution, if the manufacturers are unlikely to decrease the P_{design} for cooling.

RISE proposal is to restrict the manufacturers to market heating capacities that are higher than P_{design}. That would effectively reduce the low capacity declaration as the heating capacities used in marketing are more important than P_{design} on the Energy labeling.

Recommendations - Low capacity and colder climate declaration

RISE recommends that market restrictions are imposed to reduce the very common low capacity declaration, P_{design}, in the average climate. By restricting marketing heating capacities that are higher than P_{design} the incentive to increase P_{design} will increase and thus the Energy labeling while give the consumers of air to air and air to water heat pumps less misleading information.

Advantages

- The consumers receive information on performance that is much closer to expected performance especially in the average climate zone.
- Heat pumps manufacturers and resellers can not claim unrealistic SCOP or energy use based on low capacity declaration, in their marketing.

Disadvantages

- As SCOP likely will be lower with higher P_{design} and energy use will be higher on the Energy labelling some consumers might not take the step to buy a heat pump, especially in the implementation transition period.

Alternative test method (compensation method)

Background - RISE Experience

RISE has long experience of testing with the compensation method for air to air heat pumps in heating mode. Tests of air to air heat pumps were started in 2004. The inverter technology was rather new in 2004 and the existing standards were not feasible for testing part load performance. Instead the compensation method was used and was developed during the first set of measurements. The tests, that were done on behalf of the Swedish Energy Agency, showed the importance for the manufacturers that the products, and especially the control, needed to be adopted to work well in the Nordic cold climate. Important factors were part load control and defrosting control and performance. In all, the early tests with the compensation method, including the control of the heat pumps, challenged the manufacturers and importers of air to air heat pumps to provide well-functioning heat pumps for the Nordic market. This was one of the success factors behind the massive market growth of heat pumps in Sweden between 2005–2010 and the establishment of the mature well-functioning heat pump market in Sweden of today.

Tests continued at RISE, between 2009–2013 17 air to air heat pumps were tested and published on the Swedish Energy Agency's website. In 2016 RISE started to do tests in cooperation with the Swedish association Kyl och Värmepump Importörerna, referred to as KVI. At then, the current method with fixed frequencies was prevalent and many of the importers of air to air heat pumps requested another method that reflected the real-life performance in a better way. RISE and KVI then developed a test method based on the compensation method. From 2016–2019 8 air to air heat pumps were tested and the results have been published on KVI's webpage.

In addition to above, RISE performed tests in 2013 that resulted in a paper, Nakos et al[9]. In this paper results from tests on 4 air to air heat pumps, performed according to the harmonized standards (EN 14511:2011, EN14825:2012), were compared to test results on the same products performed according to a compensation method.

So far, RISE has minor experience from using the compensation method on brine to water and air to water heat pumps. Therefore, the analysis and recommendations within this document is mostly based on our experience from air to air heat pumps.

9. H. Nakos, C. Haglund Stignor, K. Andersson, P. Lidbom and S. Thyberg, 2014, Air to air heat pumps evaluated for Nordic climates, trends and standards, P.2.14, 11th IEA Heat Pump Conference in Montreal.

Difference between the compensation method and other alternatives/solutions and advantages/disadvantages

The compensation method enables a test that include the control system of the heat pump, in comparison with the method described in 14511:2018 and 14825:2018[10] where the control used in real life environment is overruled. Hence, the test with the method with locked frequencies may provide performance results that differ from the heat pump performance under real life conditions. And since most products available on the market today give no feedback on the actual performance, customers have no way of controlling that the performance of their heat pump is the same as promised in the marketing material.

There is also a risk that the method with locked frequencies gives small or no incentive to the manufacturers to develop efficient and stable control functions (with small temperature fluctuations) of the heat pumps. The reason is that this control function is never analyzed, which may lead to that the customer gets a product that cannot provide a high comfort climate. Tests performed recently at RISE give some indication that the control function of the heat pump of some products are very poor, resulting in a very unstable operation (and temperature) of the heat pump.

In 2014 RISE performed testing with the compensation method and compared the results with results from fixed frequency tests for air to air heat pumps. This was presented in Nakos et al.[11] and to the working group CEN TC 113 WG 7 in document N447. In the comparison it was noted that:

- For 3 of 4 heat pumps the SCOP calculated from the tested values with the compressor frequency fixed was higher (up to approx. 10%) than with the compensation method where the intrinsic control system of the heat pump adjusted the compressor capacity. Hence, the result indicates that the SCOP declaration on the energy label may not represent how the unit will work in the real-life usage.
- The results agreed well in some test points (high part loads, i.e. capacities), but differ in several test points, especially at low part load (capacities) conditions. The reason was that with the compensation method and the low capacity test conditions, it was impossible to reach the same low compressor frequencies as with the method with locked frequencies. One possible conclusion is that the control system used in real life does not allow for the low frequencies used when testing the heat pump in the lab. There is a risk of excess wear of the compressor due to lubrication problems for the lowest frequencies. Therefore, manufacturers might design the control system to not use the lowest frequencies in normal use and instead control the heat pump to use on/off operation to adjust the capacity. Anyhow, with the method with fixed frequencies there is no way of controlling whether the manufacturers are allowing for other states of operation during lab tests in comparison with real

10. * EN14825:2016/2018 describes not only a method with locked frequencies but also a compensation method to be used when frequency data is not available. However, this method is only described for air to water and brine(water) to water heat pumps and are not described in much detail. Based on our experience no-one is using this option. And seen from a market surveillance perspective, which in 2020 refers to both 14511:2013 and EN14825:2016 via the Official Journal, compensation method is not an option, since it is not described as an optional method in 14511:2013.
11. H. Nakos, C. Haglund Stignor, K. Andersson, P. Lidbom and S. Thyberg, 2014, Air to air heat pumps evaluated for Nordic climates, trends and standards, P.2.14, 11th IEA Heat Pump Conference in Montreal.

life operation.

- Since the heating capacity was different for some of the test points also the defrost pattern and its effect upon the energy performance varied between the test methods. The frost growth blocks the heat exchanger resulting in a reduced performance of the heat pump. Also, the higher the heating load of the outdoor unit heat exchanger the faster the frost growth. As mentioned above, for the low capacity test points the compensation method resulted in somewhat higher heating capacities. This gave more frost and a lower COP. For one heat pump in one test condition, the difference in frost growth was so large that heat pump had to defrost during the measurement period, something that effects the performance.

From some measurements one can see that the manufacturers define the P_{design} of their heat pumps lower than the actual maximum capacity of the products, especially for the Average climate. The test points used in 14825:2018 are based on P_{design}. And since the heat pump's actual capacity is higher than P_{design}, it is therefore difficult for the control to reach and operate stable at the low capacity test points.

From tests with the method described in EN 14511:2018 we can see that many of the heat pumps are not defrosting during the prescribed measurement time. However, we can see that frost is building up during the operation and that a defrost is necessary after the measurement time. Since the defrost affects the performance, we assume that the manufacturers are designing the heat pumps to run without having to defrost during the prescribed measurement time to achieve higher performance numbers. In real life operation, the heat pump is running continuously, resulting in frost growth and necessary defrosting. Hence, the real-life performance is likely lower than measured during the relatively short measurement time in lab.

With the compensation method there is a risk that the design of the testing rig has a non-negligible influence on the results. But based on our experience from testing where the same air to air heat pump was tested in two different indoor chambers, the influence is very small. Also varying the indoor test chamber, e.g. opening the door to the lab (with the right temperature), we have seen that for heat pumps with an unstable operation, the control still cannot provide a stable and energy efficient operation, independent of the test rig. For air to water and brine(water) to water heat pumps the influence of the test rig is currently under investigation in BAM's RR test.

With the compensation method the control of the heat pump is included in the test. Since this adds to the complexity of the test there is a risk that it is more difficult to stay within the permissible deviations allowed by the standard. A critical point is the temperature overshoot related to the power peak that often occur after the heat pump has defrosted. And for some products and test points it can be hard to reach the normal level of permissible deviations within the allowed time span of 10 minutes after the defrost is terminated. However, from our experience, these problems are most often not caused by the compensation method or the climate chamber control but by the control of the heat pump that cannot provide a stable temperature condition. Also, the problems of staying withing the permissible deviations occur in far from all the heat pumps tested.

Larger variation in temperatures, even though within the permissible deviations,

contribute to larger variations in test result. Therefore, the repeatability is somewhat worse for the compensation method than for the method with fixed frequencies. However, the repeatability is normally within the allowed measurement uncertainty described in EN14511:2018. Poor control affects the test procedure in lab leading to higher testing cost, but more important it affects the temperature comfort in real life operation. Hence, there should be a requirement on stable operation of the heat pump in the regulation. This would then force the manufacturers to design products that give good comfort to the customers, just as the early tests performed at RISE, as described above.

Using the compensation method for a heat pump with a control system unable to provide a stable operation of the heat pump may result in difficulties keeping the prescribed temperature ranges for testing and may also lead to longer stabilization times. Hence, there is a risk that the test will take longer time and be more costly to perform. But at the same time, the compensation method allows for a more automated test that could shorten the total test time and could also decrease the necessary man hours.

With the method described in EN 14511:2018 and EN 14825:2018, in case of a market surveillance test, the testing laboratory must receive input data (heat pump setting such as compressor frequencies) to be able to perform the test. In some cases, also an external control device such as a computer provided by the manufacturer is needed to perform the test. With the compensation method, neither of this input/ hardware is needed.

A noise measurement done with the compensation method risk to take longer time and be more costly than with locked frequencies. The reason is that the time it can take to reach the right operation of the heat pump. With a predefined locked frequency, the only varying factors are the temperature and humidity in the test chamber. With the compensation method, also the frequency of the compressor and perhaps also the fans (in the case of air to air and air to water heat pumps) must reach its "right" and stable operation before the test can be performed.

When could we start applying the compensation method in the regulation

As mentioned above, RISE has long experience of testing with the compensation method and have established a draft method for air to air heat pumps that could be used (maybe together with knowledge gained in the BAM RR) as a transitional method in the regulation. This would force the CEN standard working group to speed up the process include the compensation method in the standard. Testing of air to water and brine(water) to water heat pumps with the compensation method is not as well developed as for air to air units.

In our understanding, with the compensation method there is a risk that the design of the testing rig has a non-negligible influence on the test result, especially for an air to water and brine to water heat pump. The same risk might occur for an air to air heat pump but based on our experience from extensive testing the influence can be kept small. However, this influence must be investigated in Round Robin tests for all heat pump types.

Before applying the compensation method, the first step should be to establish the method fully. Our suggestion is a Round Robin test (just as the BAM tests), deep analysis of the results and revision of the method. Depending of the extension of the

revision done after the RR, maybe another smaller RR test should be performed to finalize the method.

The BAM RR for air to air heat pumps are conducted in cooling mode. The main difference in comparison to heating mode is the defrosting of the heat pump that occur at some of the test conditions in heating mode. This defrost is a critical factor for the performance of the test. Our suggestion is therefore that a RR should be done also in heating mode to provide evidence to the community of heat pumps that the method is robust and reliable also for heating mode tests.

It is difficult to give a clear indication on the times needed in the different steps towards inclusion of the compensation method in the regulation. The time needed for the standardization group is likely 18-24 months but could be longer if the group cannot agree on the details of the standard. A clear standardization mandate is necessary for completion of a new version of the standard in reasonable time, but a transitional method using the compensation method will most likely speed up the process even more.

In our opinion, the eco-design limit and class limits for energy labelling should not be changed when introducing the compensation method. In fact, the current levels are mainly based on RISE early measurements using the compensation method. Some heat pumps, mainly the ones with poor control, will have worse performance figures with the compensation method in comparison to the fixed frequency method. The results will be more in line with real life performance though.

Summary - Alternative test method

RISE has experience from the compensation method when testing air to air heat pumps since 2004. The tests, that were done on behalf of the Swedish Energy Agency, showed the importance for the manufacturers that the products, and especially the control, needed to be adopted to work well in the Nordic cold climate.

The compensation method enables a test that include the control system of the heat pump, in comparison with the method with locked frequencies where the control used in real life environment is overruled. Hence, the test with the method with locked frequencies may provide performance results that differ from the heat pump performance under real life conditions. There is also a risk that the test method described in EN 14511:2018 and EN 14825:2018 with locked frequencies gives small or no incentive to the manufacturers to develop efficient and stable control functions (with small temperature fluctuations) of the heat pumps.

In a comparison between the methods it was noted that:

- 3 of 4 heat pumps showed up to 10% lower SCOP values with the compensation method
- The largest difference in COP between the methods was for the lowest part load conditions

With the compensation method there is a risk that the design of the testing rig has a non-negligible influence on the results. However, according to our experience from air to air heat pumps testing, this influence can be kept small.

Using the compensation method may take longer time and be more costly to perform in comparison with a locked frequency test, especially for heat pumps with poor control. But at the same time, the compensation method allows for a more automated test that could shorten the total test time and could also decrease the necessary man hours. It is also important to add requirement on stable operation of the heat pump in the regulation to reduce the risk of long an expensive testing times, but more important to ensure high thermal comfort in real-life operation.

Before applying the compensation method, the first step should be to establish the method fully. Our suggestion is a Round Robin test, deep analysis of the results and revision of the method. In our opinion, the eco-design/energy marking levels should not be changed when introducing the compensation method. The reason is that the current levels are mainly based on RISE early measurements using the compensation method.

Recommendations - Alternative test method

RISE recommends the introduction of a compensation method. However, before applying the compensation method, the first step should be to establish the method fully.

Advantages

- The compensation method enables a test that include the control system of the heat pump, in comparison with the method with locked frequencies where the control used in real life environment is overruled. Hence:
  - Heat pump operation are closer to real-life operation
  - Test results are more realistic
  - Gives incentive to the manufacturers to develop efficient and stable control functions
  - Frosting and defrosting pattern in test is likely more like that in real-life operation
- Market surveillance can be performed without involvement of the manufacturer

Disadvantages

- The design of the test rig may have a non-negligible influence on the results. For air to air heat pumps this influence has been shown to be small. For air to water and brine/water to water heat pumps the influence is currently under investigation in BAM's RR test.
- Including the control in the test might lead to larger temperature variations during test which will reduce the repeatability for the compensation method in comparison to the method with fixed frequencies. However, the repeatability is high, and the results are normally within the allowed measurement uncertainty described in the standard.
- Noise measurements may take longer time to perform.

In our opinion, the eco-design/energy marking class limits should not be changed when introducing the compensation method.

Appendix A - High temperature application, 65°C supply temperature at minimum outdoor temperatures

The High Temperature Application supply temperature are very high compare to what is seen in field tests around Europe[12]. Therefore, RISE's opinion is that 65°C supply temperature is not a relevant application.

The efficiency of a heat pump is highly dependent on the temperature difference between the evaporator and the condenser. The efficiency is reduced with a larger difference. Air to water heat pumps not only have poor efficiency but can also have technical difficulties reaching 65°C supply temperature at winter conditions. The compression ratio is too great, causing high oil temperatures in the compressor and other straining conditions. Running the heat pumps at very high supply temperatures likely lead to a partly or complete switch to auxiliary heater, normally resistive heaters are used with heat pumps, at least in the Nordic countries, but other heat source are also possible of course. The auxiliary resistive heaters are usually not dimensioned to take the total heating load at winter temperatures, not having high enough capacity. Also, high amounts of auxiliary resistive heaters have a very bad impact on the electrical network using much capacity when the availability of capacity and energy normally is low.

Due to the above, to reach a well working heat pump system in an older building with unnecessarily high supply temperatures extra insulation, new energy efficient windows or other energy efficiency measures should be considered. Decreasing the energy demand with different measures is in line with the Energy Performance of Buildings Directive (EPBD). Installing additional and/or larger radiators is also a method successfully used in Sweden for several years. It doesn't reduce energy demand but decreases the needed supply temperatures in the heating system, giving a better performing heat pump system.

Overview of heat pumps on the market

Two of four major Swedish heat pump manufacturers it is not possible to set the heat pump to follow the High Temperature Application supply temperature in the Average Climate, see Figure 1 below. The green dashed line shows the Average Climate, while for heat pumps from manufacturer A the maximum possible heating curve is the top solid black line. For heat pumps from manufacturer B the maximum possible heating curve is the solid thin red line.

12. C. Haglund Stignor, T. Walfridson 2019, *Nordsyn study on air to water heat pumps in humid Nordic climate,* TemaNord 2019:502, ISBN 978-92-893-5972-6

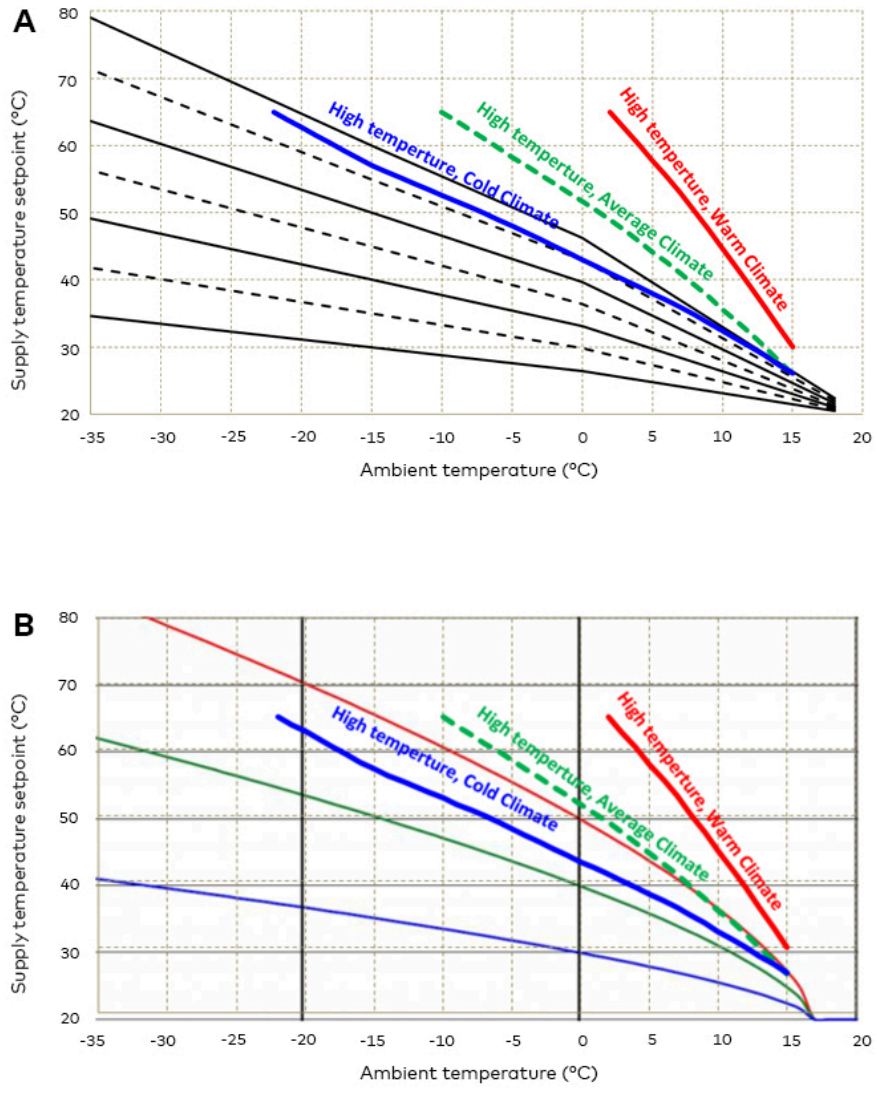


Figure 16 *Heating curves for heat pumps from manufacturer* ***A*** *and heat pumps from manufacturer* ***B****. On both these series of heat pumps it is not possible to set a heating curve corresponding to the High Temperature Application supply temperature*

On the other two major heat pump manufacturers the High Temperature Application supply temperature is very close to what is possible to set in their control systems. The green dashed line shows the Average Climate, while for the manufacturer C and D the maximum possible heating curves are shown as the top solid black lines. See Figure 17 below.

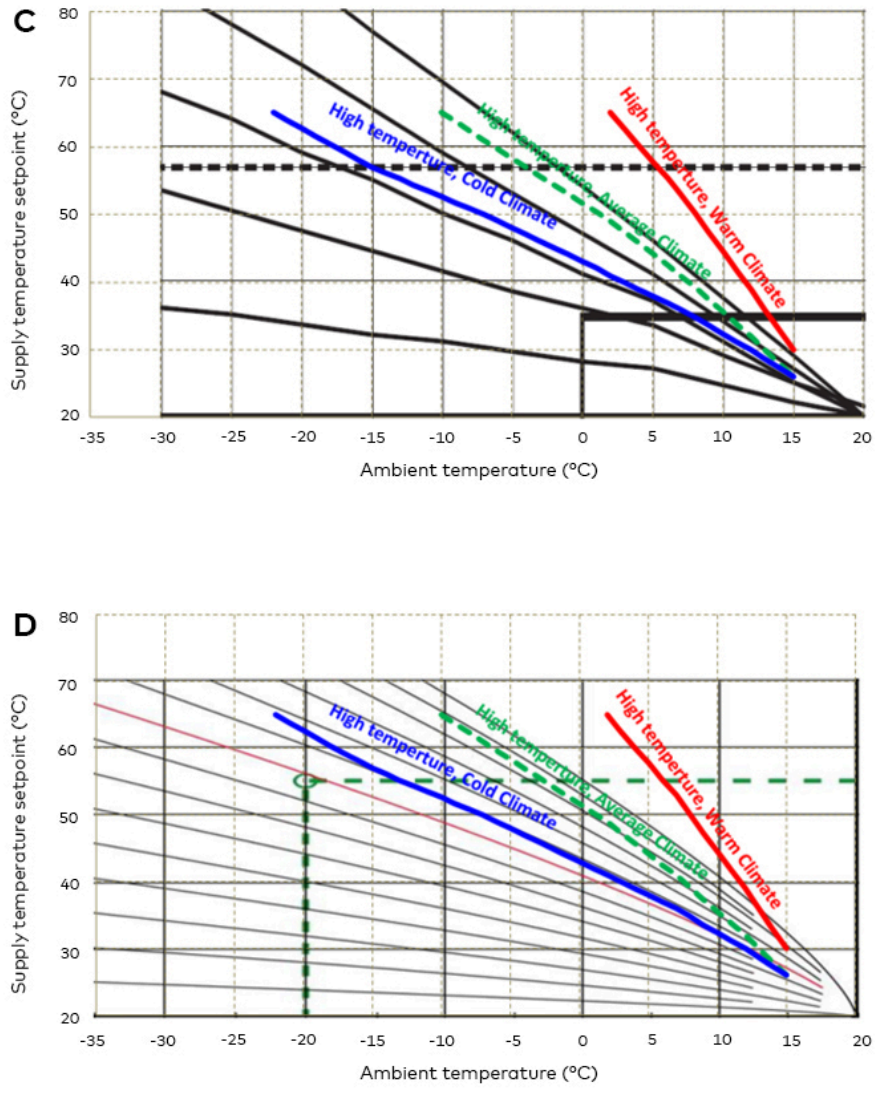


Figure 17 *Heating curves for heat pumps from manufacturer* ***C*** *and heat pumps from manufacturer* ***D****. The High Temperature Application supply temperature is on the limit of what is possible to set in their control system.*

A German heat pump manufacturer (manufacturer E), coming from the oil and gas boiler industry, has a wider set of heating curves, where all supply temperatures in the High Temperature Applications fit within their heating curves, see Figure 18 below.

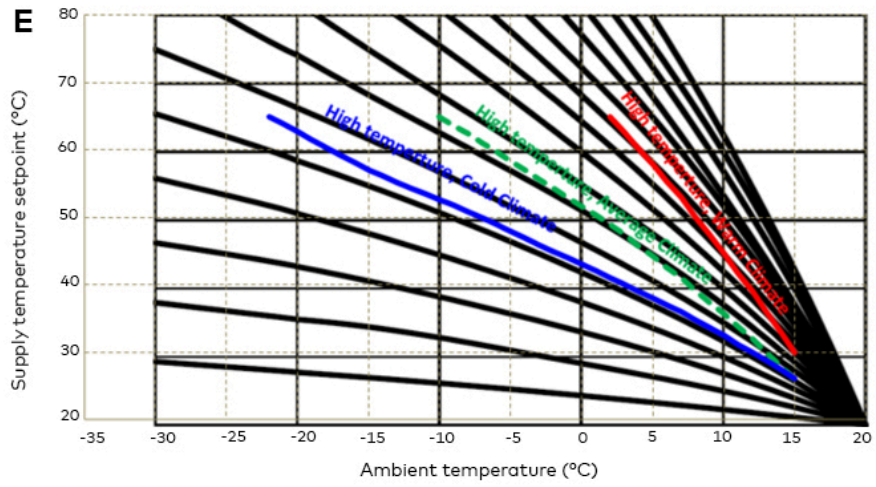


Figure 18 *Heating curves for manufacturer* ***E*** *heating equipment, including their heat pumps*

Many ground source heat pumps can deliver 65°C as their maximum supply temperature, but not all, see Figure 19 below for an example of the state of the art heat pump from manufacturer D being able to reach 65°C and the fix capacity heat pump from the same manufacturer only supplying 63°C as maximum.

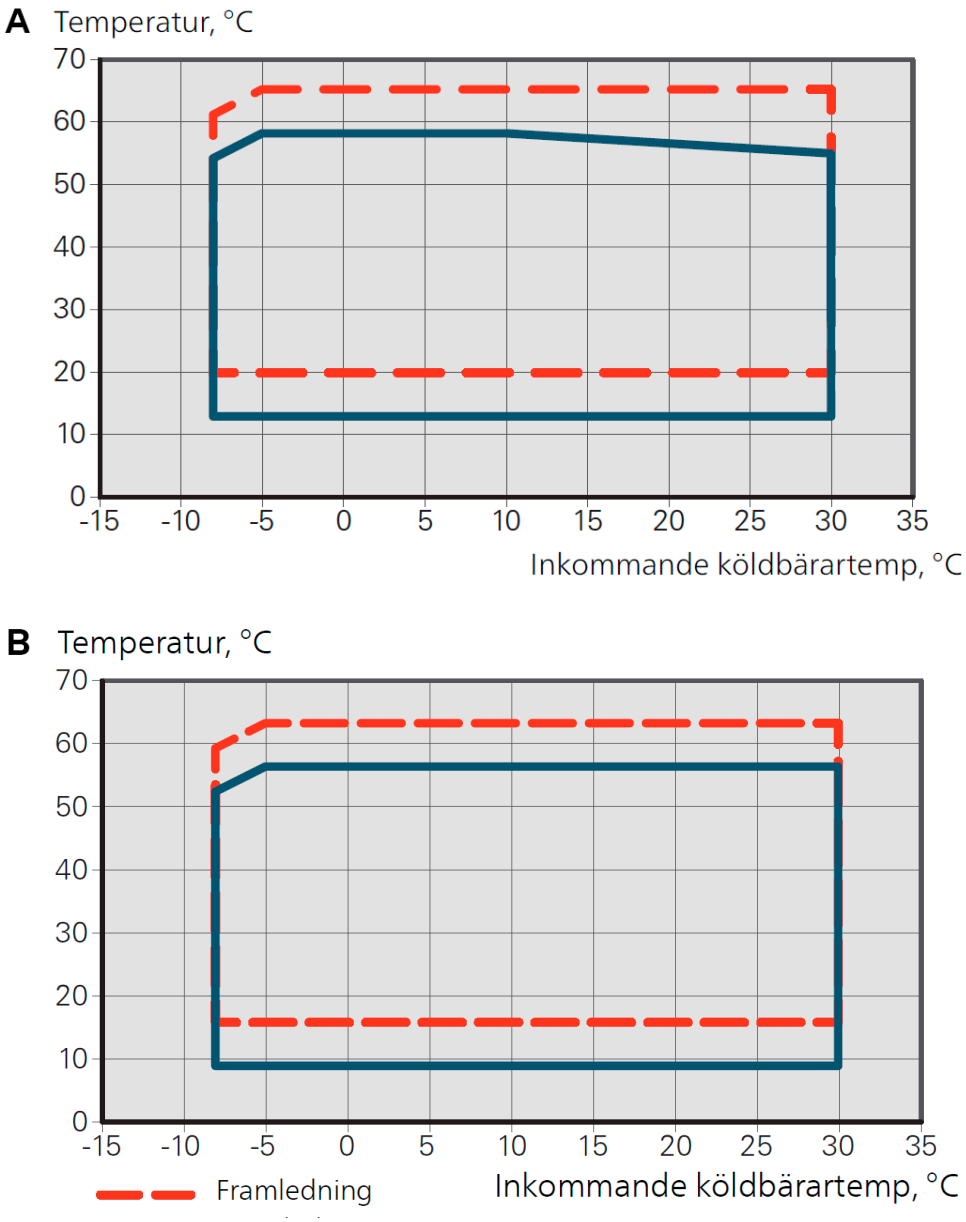


Figure 19 *Supply temperature (red dashed line) as a function of brine temperature for the state of the art ground source heat pump (**A**) and the fixed capacity heat pump (**B**). The blue solid lines represent the return temperature limits.*

Another example not being able to deliver according to the High Temperature Application is a fixed speed ground source heat pump from manufacturer B having the maximum supply temperature of 60°C. The variable speed ground source heat pumps from the same manufacturer s deliver a maximum supply temperature of 65°C

Manufacturer C's ground source heat pump series maximum supply temperature range from 62°C to 68°C.

About this publication

Alternative test methods, declaration of capacity and test of heat pumps in multiple climates zones

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